

Ch. 2 - General Theory of Experimental Modal Analysis

EXTENDED edition: 40+ pages - derivations, masterclass, banks

Whole chapter in 1 line:

Everything measured (FRF) = **POLES** (where modes live: damping + frequency) + **RESIDUES** (how strongly each mode shows at each point). Curvefitting = hunting these two. This edition PROVES every claim Ch.1 only showed.

How to use this extended edition:

Pass 1 (intuition): read S0, all [KEY] boxes, all sketches, revision. **Pass 2 (hands):** do EX-A to EX-S with a calculator - check every number. **Pass 3 (steel):** derivations + practice problems + interview bank. Exam/lab in a week? Pass 1 + formula bank + FAQ.

Legend: [KEY] key [LAB] lab [TRAP] trap [LINK] bridge [TRY]/EX do it **DERIVE** proof steps

Council crew: Dr. Meera K. (theory) - Dr. Viktor H. (lab) - Dr. Lena F. (signals) - Prof. Arjun D. (pedagogy) - Dr. Sofia M. (field) |

11 Sep 2026



(Map) Chapter roadmap - where the 40 pages go

Block	Pages do	Council owner
S0 Primer XXL	Ch.1 pictures -> Ch.2 machines; math crash kit	Arjun
2.1 Two models	FEM vs EMA, history, why theory pays	Sofia
2.2 SDOF (half the book)	Poles, S-plane, forcing, damping, vectors, Laplace, FRF, D/V/A	Meera + Lena
2.3 MDOF (the summit)	2DOF masterclass, eigen, uncoupling, FRF sums, beam, domains	Meera + Viktor
2.4 The loop	Whole EMA map, forward vs backward paths	Sofia
Banks	Formulas, symbols, FAQ, interviews, problems, revision, test, glossary	All five

[LINK] How chapters chain: Ch.2 machines -> **Ch.3** measure right (sampling/leakage) -> **Ch.4** excite right (hammer/shaker) -> **Ch.5** extract right (estimators). Theory first, craft after.

(0) Primer XXL - pictures become machines

Ch.1 picture	Ch.2 machine	You will compute
FRF peaks at fn	POLES $s = -z \omega_n + j \omega_d$	EX-B: poles from m/c/k
Imag heights trace shape	RESIDUES $\sim$ shape x shape	EX-R: shape from a column
One row/column enough	Transfer-matrix SYMMETRY	Reciprocity proof sketch
Hidden SDOFs summed	$h_{ij} = \text{SUM modal SDOFs}$	EX-M: full 2DOF FRF
Curvefit finds f,z,shape	Curvefit finds POLES+RESIDUES	Self-test + interview bank

Math crash kit (all Ch.2 needs):

- 1. Complex numbers:** $j = \sqrt{-1}$; magnitude = $\sqrt{re^2 + im^2}$; multiplying by $j = +90$ deg rotation; by $-1 = 180$ flip.
- 2. $e^{j\omega t}$:** spinning arrow at ω rad/s; derivative = multiply by $j\omega$ (velocity leads displacement 90 deg; acceleration leads 180).
- 3. dB:** $20 \log_{10}(\text{ratio})$; half power = -3 dB = divide by $\sqrt{2}$.
- 4. Matrices:** $[M]\{x\}$ couples equations; symmetric = h_{ij} -able; determinant = 0 finds eigenvalues.

Ch.1 callback quiz (60 seconds):

[TRY] 1. FRF = ? X/F in freq domain. 2. Drive-point trio? Alternate peaks/valleys, $-/+180$ phase, imag same side. 3. Reference on node? Mode vanishes. 4. Hammer vs shaker? ROW vs COLUMN. 5. Leakage cure? Windows (the tax). All green? Climb on.

(2.1) Two models, one truth - extended

	<i>FEM (analytical)</i>	<i>EMA (experimental)</i>
Starts from	Assumed M, K layout + BCs	Measured force + response
Engine	EIGENSOLUTION of big matrices	CURVEFITTING of FRFs
Gives	Freqs + shapes (no damping!)	f, z, shapes (WITH damping)
Strength	Design-stage what-ifs, zero hardware	Ground truth incl. joints/damping
Weakness	Wrong assumptions/BCs = fiction	Bad FRFs = fiction with error bars
Costs	Engineer time + compute	Sensors, analyzer, rig, lab days

Why FEM needs EMA (Sofia's war story):

A bracket FEM said 1st mode 210 Hz; test said 164 Hz. Culprit: bolted joint modeled as welded (infinite stiffness). Fix: bushing elements tuned to test, error 210 → 166 Hz. **[KEY]** Joints, welds, gaskets, preload: FEM guesses, EMA KNOWS. Damping is never predicted – only measured.

<i>Era</i>	<i>What changed for modal testing</i>
1980s	FFT analyzers + IMAC conferences; EMA becomes routine
1990s	MIMO acquisition + better estimators; big structures testable
2000s	OMA (no-force methods); laser vibrometry spreads
2010s+	High-channel counts, model updating + UQ pipelines

Book's promise for Ch.2: NOT every derivation – the equations that change how you TEST, with proofs only where they protect you from mistakes. **[TRAP]** Vibration textbooks derive; this chapter ARMS. Different goal – read accordingly.

(2.2A) SDOF 1: equation, poles, S-plane

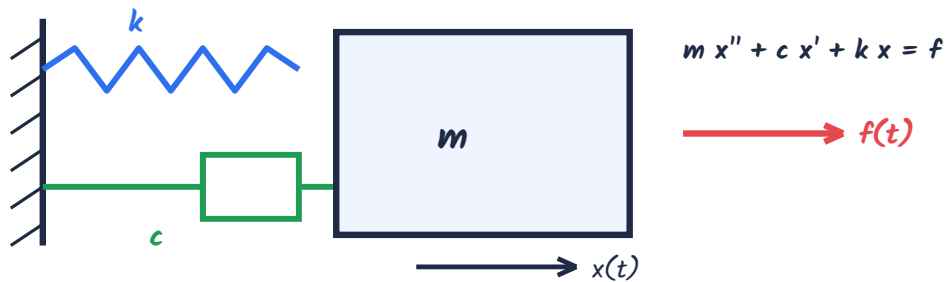
The equation + its honest assumptions:

$$m \ddot{x} + c \dot{x} + k x = f(t) \text{ (linear, time-invariant)}$$

<i>Assumption</i>	<i>Means</i>	<i>Breaks when</i>
Lumped m	Mass acts at a point	Distributed inertia matters (beams at high f)
Linear k x	Stiffness constant	Gaps, hardening springs, buckling
Viscous c $\dot{x}$	Damping ~ velocity	Friction (Coulomb), joints, air pumping
Time-invariant	m/c/k frozen	Fuel burn, heat, wear, opening cracks

[TRAP] Everything in Ch.2 dies with nonlinearity – but note the mercy: lightly nonlinear structures still give USEFUL linear fits per amplitude. Later chapters teach detecting the lie (distorted Nyquist, level-dependent freqs).

my sketch: the SDOF machine



EX-A: quarter-car suspension (feel real numbers):

$m = 400 \text{ kg}$, $k = 30 \text{ kN/m}$, $c = 2 \text{ kN s/m}$. $\omega_n = \sqrt{30000/400} = 8.66 \text{ rad/s} \rightarrow f_n = 1.38 \text{ Hz}$ (that floaty car feel!). $cc = 2 \cdot 400 \cdot 8.66 = 6928$. $z = 2000/6928 = 0.29$ - comfy AND controlled. Static sag under weight: $400 \cdot 9.81/30000 = 13 \text{ cm}$. Now you know why speedbreakers at 1.4 Hz bounce hurt.

DERIVE: characteristic equation to poles (follow with pencil):

- > Step 1: free vibration, $f = 0$: $m x'' + c x' + k x = 0$.
- > Step 2: try $x = X e^{st}$: $x' = sX e^{st}$, $x'' = s^2 X e^{st}$.
- > Step 3: factor $X e^{st}$ [never zero] -> $m s^2 + c s + k = 0$. THE characteristic equation.
- > Step 4: quadratic formula: $s = [-c \pm \sqrt{c^2 - 4mk}] / 2m$.
- > Step 5: underdamped ($c^2 < 4mk$): sqrt gives j : $s = -c/2m \pm j \sqrt{4mk - c^2}/2m$.
- > Step 6: name $c/2m = z \omega_n$ and the imag part ω_d -> $s = -z \omega_n \pm j \omega_d$. DONE - poles derived, not memorized.

$$\omega_n = \sqrt{k/m}, cc = 2 m \omega_n, z = c/cc, \omega_d = \omega_n \sqrt{1-z^2}$$

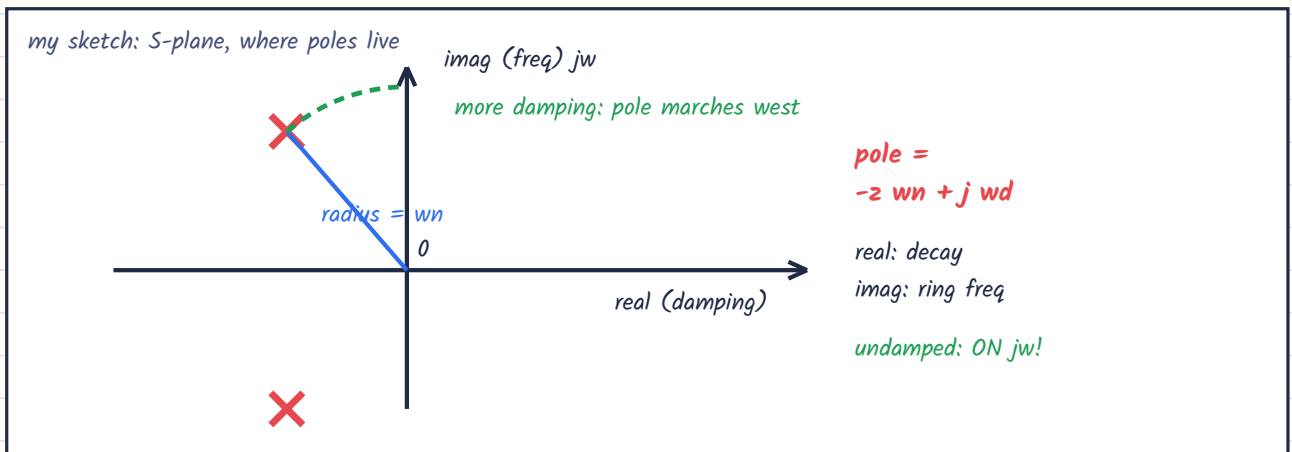
EX-B: pole drill (three fates - compute, then check):

- (a) $m=0.5$, $k=2000$, $c=2$: $\omega_n=63.2$, $cc=63.2$, $z=0.032$, poles $-2 \pm j63.2$ (rings ~10 Hz, light).
- (b) $m=10$, $k=40000$, $c=400$: $\omega_n=63.2$, $cc=1265$, $z=0.316$, $\omega_d=60.0$, poles $-20 \pm j60$ (heavy - shock absorber territory).
- (c) $c = cc$ exactly: **double real pole** $-\omega_n$ - critical, the border crossing.

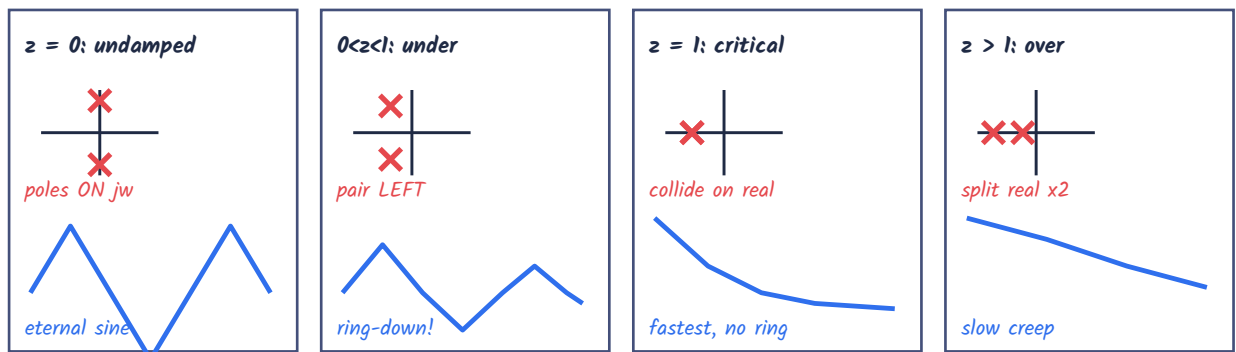
Pole anatomy - tattoo this:

[KEY] Real ($-z \omega_n$) = decay rate. Imag (ω_d) = ring freq. **Conjugate pairs** always (real system!). $z < 0.1$: $\omega_d \sim \omega_n$ (error under 0.5%). ω_n NEVER depends on damping - only ω_d does.

S-plane deep map:



my sketch: pole migration vs damping (4 fates)



-> Read the 4 fates: **undamped** ON jw (eternal sine); **under** left-shifted pair (ring-down + sharp peak); **critical** merged on real axis (fastest return, zero overshoot); **over** split real pair (slow creep).

[KEY] Radius origin-to-pole = w_n , ALWAYS. Damping rotates the pole west along the circle; stiffness scales the circle. **[LINK]** MAP reading: cities (poles) west = damped, north = stiff. Your FRF = the view driving up the jw highway.

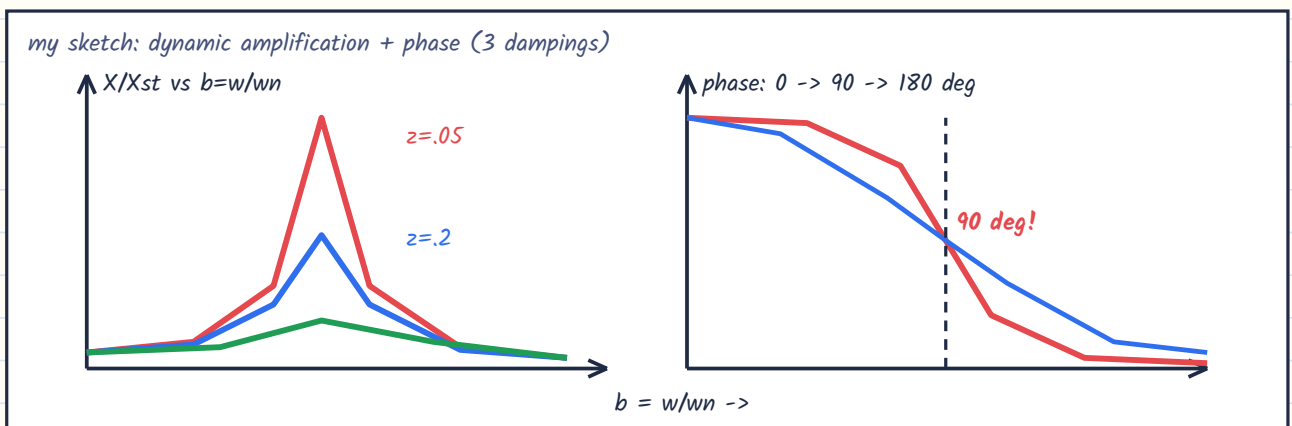
[TRY] EX-C: sketch S-planes for EX-B (a) and (b) - same radius 63.2, but (b) sits MUCH further west. Which rings longer? (a), by 10x - decay time $1/(z w_n)$: 0.5 s vs 0.05 s.

Forced response: amplification DERIVED (not gifted):

-> Step 1: $f = F e^{j\omega t}$, guess $x = X e^{j\omega t} \rightarrow (-m \omega^2 + j c \omega + k) X = F$.

-> Step 2: $X/F = 1 / [(k - m \omega^2) + j c \omega]$. Divide top+bottom by k ; set $b = \omega/w_n$, $c \omega/k = 2 z b$.

-> Step 3: $X/X_{static} = 1 / [(1-b^2) + j 2 z b]$. Magnitude: $1/\sqrt{(1-b^2)^2 + (2zb)^2}$. Phase: $\text{atan2}(2zb, 1-b^2)$.



EX-D: amplification table (compute ONE row, trust the rest):

$z=0.02$: $b=0.9 \rightarrow 5.17$; $b=1 \rightarrow 25.0$; $b=1.1 \rightarrow 4.66$.

$z=0.05$: $4.76 / 10.0 / 4.22$.

$z=0.10$: $3.82 / 5.0 / 3.29$.

[KEY] Moral: peak height $\sim 1/(2z)$ - doubling damping HALVES the peak. Off-resonance ($b=0.9/1.1$) damping barely matters - stiffness/mass rule there.

Phase: the 90-degree signature:

[KEY] Lag 0 deg (static, moves WITH force) $\rightarrow$ 90 deg AT resonance $\rightarrow$ 180 deg (moves AGAINST force).

[LAB] Peaks lie (noise/leakage); 90 deg rarely lies. No 90 crossing = no resonance, whatever the bump says. - Lena

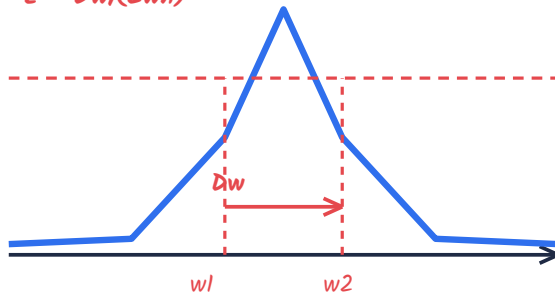
[TRY] EX-E: $z=0.02$, $b=0.9$: phase = $\text{atan2}(0.036, 0.19) = 10.7$ deg. $b=1.1$: $\text{atan2}(0.044, -0.21) = 168$ deg. Feel the snap across resonance!

(2.2B) SDOF II: damping, vectors, Laplace, FRF

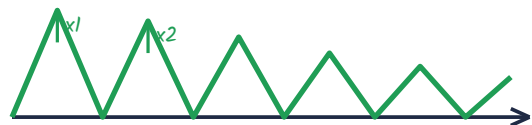
Half-power: DERIVED in 4 steps (small- z approx):

my sketch: half-power + log-dec (damping quickies)

$$z = D\omega/(2\omega_n)$$



$$\text{log-dec: } d = \ln(x1/x2)$$



$\rightarrow$ Step 1: peak $|H| \sim 1/(2z)$ at $b=1$. Half-power: $|H|_2 = \text{half of peak} \rightarrow |H| = \text{peak}/\sqrt{2}$ (-3 dB).

$\rightarrow$ Step 2: solve $1/[(1-b^2)^2 + (2zb)^2] = (1/2)(1/(2z)^2)$ near $b=1$: write $b = 1+e$, keep first order: $(2e)^2 + (2z)^2 = 2(2z)^2$.

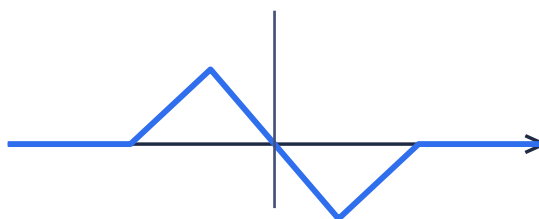
$\rightarrow$ Step 3: $e^2 = z^2 \rightarrow e = \pm z \rightarrow b_{1,2} = 1 \pm z$.

$\rightarrow$ Step 4: bandwidth $D_b = 2z \rightarrow z = (w_2 - w_1)/(2\omega_n)$. Wider peak = more damping. Q.E.D.

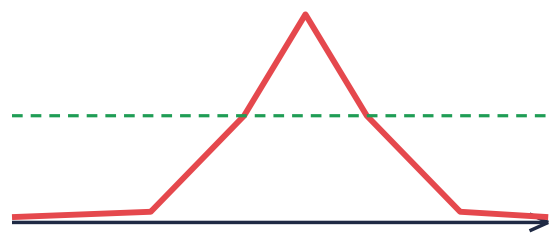
$$z \sim D\omega/(2\omega_n) = (f_2 - f_1)/(2f_n), Q = 1/(2z)$$

my sketch: half-power on co-quad (real + imag agree!)

REAL: 0 at reso, peaks = hp pts



IMAG: peak; half pts = hp



same w_1, w_2 both sides!

-> Co-quad witnesses: **REAL peaks sit exactly at hp points, REAL = 0 at resonance; IMAG peaks at resonance, half-height at hp.** Nyquist: hp points **90 deg around the circle** from resonance. Three portraits, one verdict.

[TRY] EX-F: peak 100 Hz, hp at 99/101 -> $z = 2/200 = 1\%$, $Q = 50$. Peak 100 Hz, hp at 96/104 -> $z = 4\%$, $Q = 12.5$. Same fn, 4x the damping - SEE the width!

Log-decrement: **DERIVED + measured right:**

-> Free ring: $x(t) = A e^{-(z \omega_n t)} \sin(\omega_d t)$. One period later ($T_d = 2\pi/\omega_d$): ratio $x_1/x_2 = e^{(z \omega_n T_d)} = e^{(2\pi z / \sqrt{1-z^2})} \sim e^{(2\pi z)}$. Take ln: **$d = \ln(x_1/x_2) \sim 2\pi z$** .

-> Pro move: average over n cycles: $d = \ln(x_1/x_{n+1})/n$ - kills reading noise. Plot $\ln(\text{peak})$ vs cycle: **straight line = linear viscous**; curving = amplitude-dependent (nonlinear!) damping.

$$d = \ln(x_1/x_2) \sim 2\pi z \text{ (small } z, \text{ single mode!)}$$

EX-G: ring-down table (single mode, 5 Hz):

Peaks: 10.0, 8.2, 6.7, 5.5 mm. $d_1 = \ln(10/8.2) = 0.198$; $d_2 = \ln(8.2/6.7) = 0.202$; $d_3 = \ln(6.7/5.5) = 0.198$.

Steady ~ 0.20 -> $z = 0.20/(2\pi) = 3.2\%$. **[TRAP]** If d DRIFTED (0.2, 0.25, 0.35...) - suspect friction or multi-mode beating, NOT viscous damping.

Museum verdict (read twice):

[TRAP] Half-power + log-dec: exam-favorite, history-honored - but real responses are multi-mode, so they mislead. **[LAB] Pros fit poles+residues (least squares)**. Quickies for sanity + interviews; estimators for paychecks. - Viktor

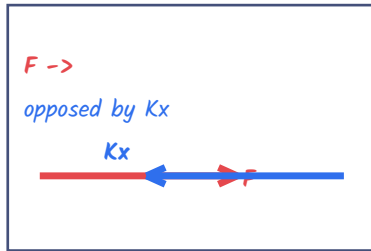
Language	Formula	EX-G value
Damping ratio z	c/cc	3.2%
Q factor	$1/(2z)$	~ 16
Bandwidth	$\Delta\omega = 2 z \omega_n$	2.0 rad/s
Log-dec d	$2\pi z$	0.20
Loss per cycle	$\sim 4\pi z$ (small z)	$\sim 40\%$ amplitude/cycle? NO: $\sim 18\%$ ($e^{-0.2}$)

[TRY] EX-H: $z = 1\% \leftrightarrow Q = 50$, $d = 0.0628$. $z = 5\% \leftrightarrow Q = 10$, $d = 0.314$. Memorize the pairs (1/50), (2/25), (5/10) - fluent damping in 3 numbers.

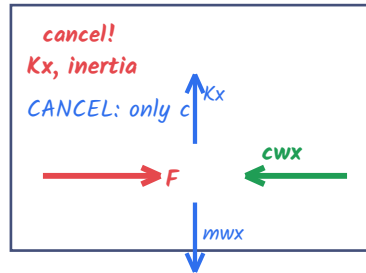
Force balance: the vector war, with math:

my sketch: who balances F ? (force vectors)

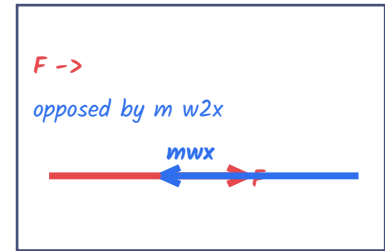
$w \ll \omega_n$: STATIC-like



$w = \omega_n$: DAMPING only!



$w \gg \omega_n$: MASS rules



-> Phasors (all at ω): elastic KX at 0 deg; damping $c \omega X$ at +90; inertia $m \omega^2 X$ at 180. Sum + $F = 0$ ALWAYS.

-> Below reso: KX huge vs $m \omega^2 X \rightarrow F \sim KX$ (static-like). Above: $m \omega^2 X$ wins $\rightarrow F \sim m \omega^2 X$. AT reso: $KX = m \omega^2 X$ exactly (that IS $\omega_n^2 = k/m$!) $\rightarrow$ they annihilate $\rightarrow F = c \omega X$ alone.

Deepest insight + numbers:

[KEY] Resonance peak measures DAMPING and nothing else - because stiffness and inertia signed a mutual destruction pact.

EX3: $m=2$, $k=8000$ at $\omega_n=63.2$, $X=1\text{mm}$: $KX = 8 \text{ N UP}$, inertia = 8 N DOWN , damping = 0.5 N . 8 vs 0.5 : who is holding the bridge?

EX-1: same rig at $\omega = 20$ ($b=0.32$): $KX = 8$, inertia = $2 \cdot 400 \cdot 0.001 = 0.8$, damping = $8 \cdot 20 \cdot 0.001 = 0.16$. $F \sim 8 - 0.8 = 7.2 \text{ N}$: 90% stiffness. AT $\omega=200$: inertia = 80 N dominates: mass world.



Laplace: 3 costumes, zero new physics:

Form	Looks like	Best for
Polynomial	$H = 1/(m s^2 + c s + k)$	Seeing the equation
Pole-zero	$H = (1/m)/[(s-p_1)(s-p_2)]$	Seeing the poles
Partial fraction	$H = A_1/(s-p_1) + A_2/(s-p_2)$	TESTING: poles + residues!

my sketch: cover-up method for residues (SDOF)

$$H(s) = (1/m) / [(s-p_1)(s-p_2)]$$

cover $(s-p_1)$!



$$\text{set } s = p_1 \rightarrow A_1 = (1/m) / (p_1 - p_2)$$

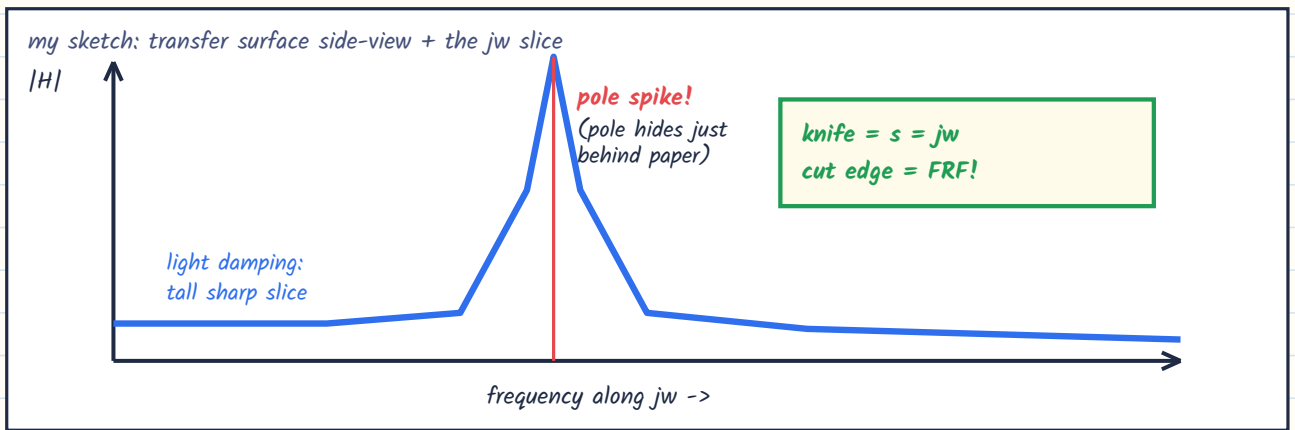


$$p_1 - p_2 = j 2 \omega_d, \text{ so } A_1 = 1 / (j 2 m \omega_d)$$

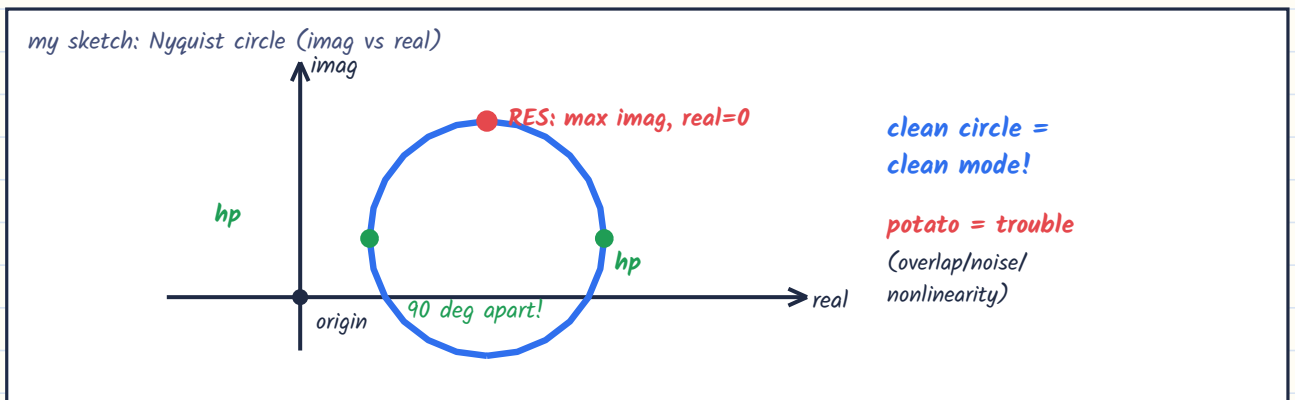
Cover-up = limit trick: $A_1 = \lim_{s \rightarrow p_1} (s-p_1) H(s)$ as $s \rightarrow p_1$. For SDOF: $A_1 = (1/m)/(p_1-p_2) = 1/(j 2 m \omega_d)$ (pure imaginary!). **[KEY]** Pole + residue = DNA: rebuild $H(s)$ everywhere + FRF at every f .

[TRY] EX-J: EX3 rig ($m=2$, $\omega_d=63.2$): $A_1 = 1/(j \cdot 2 \cdot 2 \cdot 63.2) = -j 0.00198$. Tiny? It multiplies $1/(j\omega - p_1)$ which is HUGE near resonance - product = the giant peak.

FRF = slice at $s = j\omega$ (see it!):



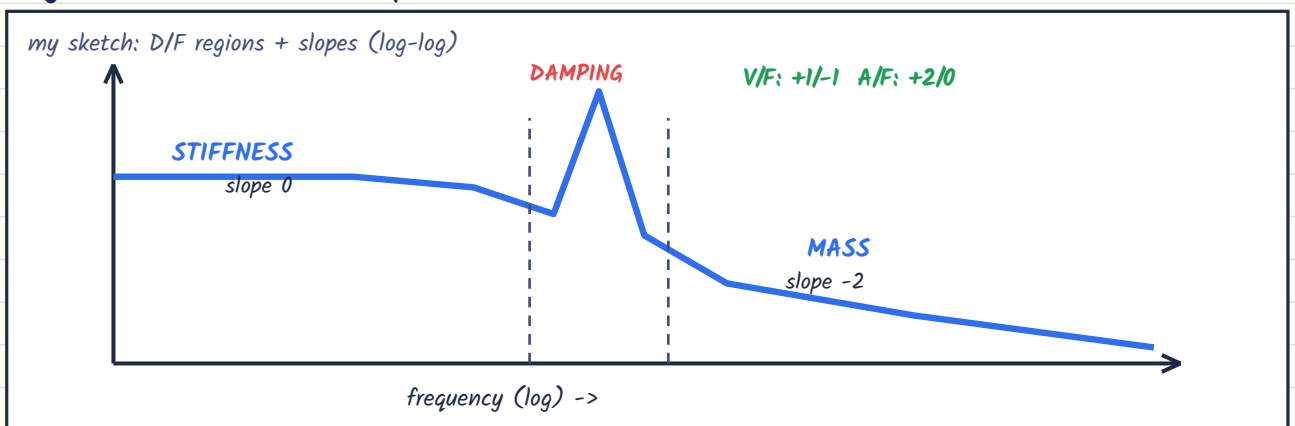
Portrait	Shows	Resonance reads as
Bode (mag + phase)	Peak + 90 deg, 180 total flip	Peak + 90 crossing
Co-quad (real + imag)	Real crosses ZERO, imag PEAKS	Imag peak, real = 0
Nyquist (imag vs real)	Near-CIRCLE	Opposite origin; hp 90 deg around



[LAB] Circle = clean mode. POTATO = overlap/noise/nonlinearity. Sweep direction + speed also talks: data points bunch AT resonance (fast phase change) - sparse elsewhere. - Lena

[TRY] EX-K: Nyquist top point reads (0.00, 2.5). hp points at (+-1.25, 1.25). Peak imag = 2.5, hp at half height 1.25 with $|real| = 1.25$: textbook SDOF. If hp reals were 0.4/2.1 (asymmetric) - neighbor mode leaking in!

Regions + D/N/A: the slope detective kit:



Name	Symbol	From D/F	Stiff slope	Mass slope
Compliance	D/F	-	0 (flat)	-2

Name	Symbol	From D/F	Stiff slope	Mass slope
Mobility	V/F	$\times j\omega$	+1	-1
Inertance	A/F	$\times -\omega^2$	+2	0 (flat)

[KEY] Each $j\omega$ multiply: +1 to BOTH slopes, +90 phase. A/F = $-\omega^2$ (D/F): 180 apart, ω^2 scaled. **[LAB]** Unlabeled plot? Far slopes (0,-2)=D, (+1,-1)=V, (+2,0)=A. Seconds, not meetings.

EX-L: slope detective quiz (answers inline, think first!):

1. Flat left, -2 right $\rightarrow$ **D/F**. 2. +1 left, -1 right $\rightarrow$ **V/F**. 3. +2 left, flat right $\rightarrow$ **A/F** (your hammer+accel daily driver). 4. Someone's 'A/F' shows -2 far right $\rightarrow$ **MISLABELED: it is D/F**. Caught!

EX-M0: convert across forms (50 Hz, $D = 2e-6$ m/N at -170 deg):

$\omega = 314$. V/F = $j\omega \times D$: mag $314 \times 2e-6 = 6.28e-4$, phase $-170+90 = -80$ deg. A/F = $-\omega^2 \times D$: mag $98696 \times 2e-6 = 0.197$, phase $-170+180 = +10$ deg. One measurement, three costumes.

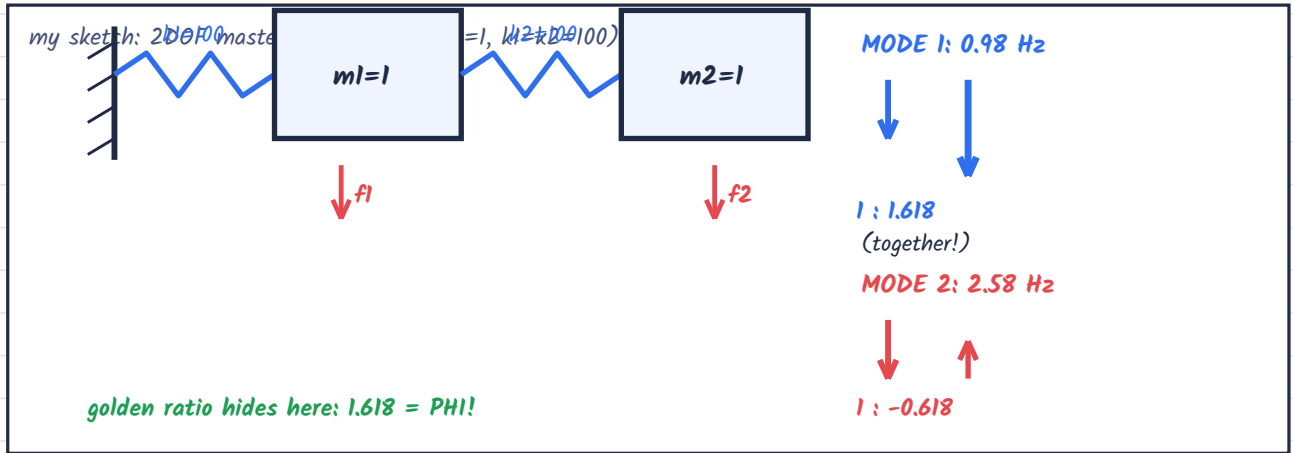
Nuance most books bury: THREE different peaks!

[TRAP] D/F peaks at **wd-ish BELOW ω_n** ; V/F peaks **EXACTLY at ω_n** ; A/F peaks **ABOVE ω_n** . Light damping: all $\sim \omega_n$ (relax). Heavy damping ($\zeta=0.3$): D peaks $\sim 0.90 \omega_n$, A peaks $\sim 1.10 \omega_n$ - **20% apart!** Always say WHICH FRF your 'resonance' came from.



(2.3A) MDOF 1: 2DOF masterclass - solved by hand

The rig: $m_1 = m_2 = 1 \text{ kg}$, $k_1 = k_2 = 100 \text{ N/m}$ (m_2 free at the end). Small enough to solve on paper, rich enough to show EVERYTHING. Follow with a calculator - trust nothing.



Step 1 - equations of motion (force balance per mass):

$$m_1 x_1'' + (k_1 + k_2) x_1 - k_2 x_2 = f_1$$

$$m_2 x_2'' - k_2 x_1 + k_2 x_2 = f_2$$

In matrices: $M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $K = \begin{bmatrix} 200 & -100 \\ -100 & 100 \end{bmatrix}$. [KEY] Off-diagonals (-100) = the coupling springs talking. Both matrices SYMMETRIC - reciprocity is already baked in.

Step 2 - eigenvalues (free vibration, assume $e^{j\omega t}$):

$$\rightarrow \det(K - L M) = (200 - L)(100 - L) - 10000 = L^2 - 300L + 10000 = 0.$$

$$\rightarrow L = [300 \pm \sqrt{90000 - 40000}]/2 = [300 \pm 223.6]/2.$$

$$L_1 = 38.2 \rightarrow \omega_1 = 6.18 \text{ rad/s } (f_1 = 0.98 \text{ Hz})$$

$$L_2 = 261.8 \rightarrow \omega_2 = 16.18 \text{ rad/s } (f_2 = 2.58 \text{ Hz})$$

Step 3 - eigenvectors (shapes): back-substitute each L :

$$\rightarrow \text{Row 1: } (200 - L) u_1 - 100 u_2 = 0 \rightarrow u_2/u_1 = (200 - L)/100.$$

$$\rightarrow \text{Mode 1: } (200 - 38.2)/100 = 1.618 \rightarrow u_1 = [1, 1.618] \text{ (together, mass 2 swings 62\% MORE).}$$

$$\rightarrow \text{Mode 2: } (200 - 261.8)/100 = -0.618 \rightarrow u_2 = [1, -0.618] \text{ (opposite, mass 2 swings less).}$$

Golden ratio Easter egg:

[KEY] 1.618 = PHI, and 0.618 = 1/PHI. Not a coincidence of YOUR rig only - symmetric 2DOF systems breed golden ratios. Nature signs her work.

Step 4 - orthogonality CHECK (prove it, don't quote it):

$$u_1(T)^T M u_2 = 1 \times 1 + 1.618 \times (-0.618) = 1 - 1.000 = 0. \text{ VERIFIED.}$$

If this were NOT ~ 0 , your eigensolver (or arithmetic) lied. [LAB] Pros run this as *pseudo-orthogonality*: test shapes vs FEM shapes through FEM mass matrix. Off-diagonal ~ 0 = shapes agree; big values = somebody is

wrong.

Step 5 - modal masses + stiffnesses:

Mode	$m_i = u(T)Mu$	$k_i = u(T)Ku$	Check: $\omega_i^2 \times m_i$
1	$1 + 1.618^2 = 3.618$	138.2	$38.2 \times 3.618 = 138.2$ OK
2	$1 + 0.618^2 = 1.382$	361.8	$261.8 \times 1.382 = 361.8$ OK

$k_i = \omega_i^2 m_i$ MUST hold (it is the same eigenfact, weighed). Two independent computations agreeing = you are safe.

Step 6 - mass-normalize (divide by $\sqrt{m_i}$):

$$PHI1 = [0.526, 0.851], PHI2 = [0.851, -0.526]$$

Unity modal mass ($PHI(T) M PHI = 1$) is THE standard scaling: residues become clean shape-products. **[TRAP]** Unscaled shapes ([1, 1.618]) show PATTERN only; scaled shapes give correct RESPONSE levels. Mixing them = wrong amplitudes!

Step 7 - residues (shape-product table - REAL numbers):

	$h11$	$h12 = h21$	$h22$
Mode 1 (A)	0.276	+0.447	0.724
Mode 2 (A)	0.724	-0.447	0.276

[KEY] Reciprocity VISIBLE: $A12 = A21$ in EVERY mode (+0.447 / -0.447). Residue SIGN = peak direction: mode 2's $h12$ imag peak points DOWN. Negative peaks are not errors!

Step 8 - STATICS verify DYNAMICS (the mic-drop check):

$$h11(0) = 0.276/38.2 + 0.724/261.8 = 0.00724 + 0.00276 = 0.0100$$

Push $m1$ statically: ONLY $k1$ stretches ($m2$ rides free) $\rightarrow x1 = F/100 = 0.01$. Modal sum says 0.0100. EXACT.

[KEY] Two alien math worlds agree to 4 decimals - the theory is TRUE, not just pretty.

$\rightarrow$ Mode 1 alone gives 0.00724 (72%); mode 2 contributes 28% as **residual stiffness**. Truncate mode 2 and you lose over a quarter of static answer - residuals MATTER (1.7 callback!).

Step 9 - full FRF near resonance ($z1 = 2\%$, $z2 = 3\%$, $w = 6$):

$\rightarrow$ Mode 1 denom: $(38.2-36) + j(2 \times 0.02 \times 6.18 \times 6) = 2.2 + j1.48$; term = $0.276/2.657$ at -34 deg = 0.104 at -34 deg.

$\rightarrow$ Mode 2 denom: $(261.8-36) + j(2 \times 0.03 \times 16.18 \times 6) = 225.8 + j5.83$; term = 0.0032 at -1.5 deg (negligible!).

$$h11(6 \text{ rad/s}) \sim 0.089 - j0.058 (|.| \sim 0.107, \text{ angle } -33 \text{ deg})$$

[KEY] At 6 rad/s (3% below mode 1): response is 97% MODE 1. Off-resonance modes answer quasi-statically - small, real, in-phase. THIS is why truncation + residuals works.



Masterclass graduation:

You just: wrote EOMs, solved eigen by hand, verified orthogonality, weighed modal m/k , normalized shapes, built residues, verified against statics, and evaluated a damped FRF. That is literally Ch.2's whole engine, fingerprinted. **[TRY]** Victory lap: recompute $h_{22}(0) = 0.724/38.2 + 0.276/261.8 = 0.0200$. Static check: push m_2 : k_1 AND k_2 in series $\rightarrow x_2 = F(1/100+1/100) = 0.02$. Again exact. Cry a little.

(2.3B) MDOF II: eigen to FRF sums

Eigensolution, conceptually (the book's walk-through):

- $\rightarrow$ Step 1: $\det(K - \omega^2 M) = 0 \rightarrow$ polynomial $\rightarrow$ roots = eigenvalues (ω^2). Small rigs: Jacobi/Givens/Householder (direct, ALL modes). Giant FEM: subspace/Lanczos (iterative, LOWEST modes only - the ones that matter).
- $\rightarrow$ Step 2: plug L_1 back, solve (Crout/Cholesky/LDL) $\rightarrow$ vector = shape 1. Repeat. (EX-M did this by hand - same moves!)
- $\rightarrow$ Step 3: physics at each root: elastic = inertial forces (DYNAMIC EQUILIBRIUM), nodes with equal $+/ -$ lobes.

Solver family	Gets	Use when
Direct (Jacobi, Givens, Householder)	ALL eigenpairs, exact-ish	Small matrices, teaching, checks
Iterative (subspace, Lanczos)	Lowest m modes	100k+ DOF FEM (cars, planes)
Root polish (Newton/Secant)	Refines det roots	Inside hybrid codes

EX-N: 2DOF intuition (say it in words):

Mode 1 (0.98 Hz): masses TOGETHER, middle spring lazy $\rightarrow$ soft $\rightarrow$ LOW f . Mode 2 (2.58 Hz): masses OPPOSED, middle spring stretched double $\rightarrow$ stiff $\rightarrow$ HIGH f . **[KEY]** Every MDOF mode = an independent dance move; every real motion = a mix. (EX-M proved the numbers; this is the story.)

DERIVE: orthogonality in 5 lines (the book p145-146):

- $\rightarrow$ Line 1: $K u_i = \omega_i^2 M u_i$ (eigen- i). Line 2: pre-multiply by $u_j(T)$: $u_j(T) K u_i = \omega_i^2 u_j(T) M u_i$.
- $\rightarrow$ Line 3: same with i, j swapped: $u_i(T) K u_j = \omega_j^2 u_i(T) M u_j$.
- $\rightarrow$ Line 4: transpose line 3 (K, M symmetric!): $u_j(T) K u_i = \omega_j^2 u_j(T) M u_i$.
- $\rightarrow$ Line 5: subtract from line 2: $0 = (\omega_i^2 - \omega_j^2) u_j(T) M u_i \rightarrow \omega_i \neq \omega_j$ forces **$u_j(T) M u_i = 0$** . SAME for K . DONE.

$$i \neq j: u_j(T) M u_i = 0, u_j(T) K u_i = 0. i=j: m_i, k_i \text{ (modal } m, k).$$

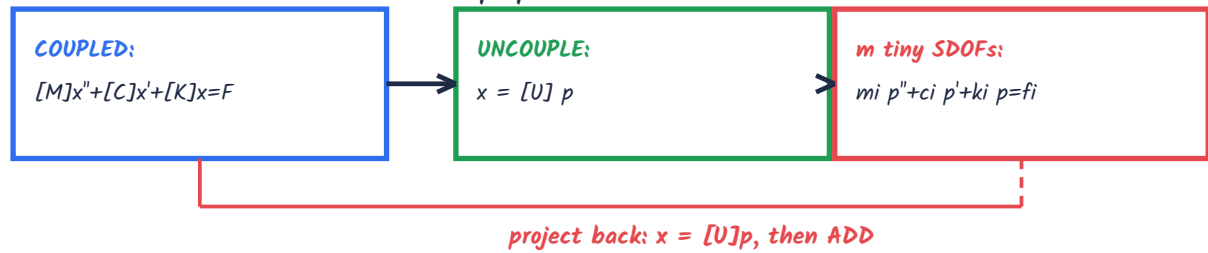
[TRAP] SYMMETRY did all the work (line 4). No symmetry (gyros, control feedback) = no clean orthogonality = complex modes (preview, not Ch.2).

Modal transformation: the uncoupling machine:

$$\{x\} = [U]\{p\} \rightarrow m_i p'' + c_i p' + k_i p = f_i = u_i(T)\{F\}$$

my sketch: modal transformation round-trip

modal superposition!



-> Why it works: pre-multiply EOM by $[U](T)$: $[U](T)[M][U] = \text{diag}(m_i)$ and $[U](T)[K][U] = \text{diag}(k_i)$ BY orthogonality. Damping needs **proportional C** ($aM + bK$) to join the diagonal party.

-> Modal force $f_i = u_i(T) F$: dot product = how **ALIGNED** the push is with shape i . Push at a node -> $f_i = 0$ -> mode sleeps. (Node rule, third birth certificate!)

[TRY] EX-P: EX-M rig, $F = [1, 0]$ (push m_1 only). Modal forces (unnormalized u): $f_1 = 1 \times 1 + 1.618 \times 0 = 1.0$; $f_2 = 1 \times 1 + (-0.618) \times 0 = 1.0$. Push m_2 only, $F = [0, 1]$: $f_1 = 1.618$ (mode 1 LOVES tip pushes), $f_2 = -0.618$. Same newton, different music!

Truncation + residuals: why $m \ll n$ works:

-> Keep modes to ~ 1.5 - $2 \times$ your max frequency of interest. Higher modes answer quasi-STATICALLY (they never resonate in-band) -> fold them into **residual stiffness/mass terms** (Ch.1 1.7 callback: the curvefitter's 'residuals' input!).

-> EX-M receipt: mode-1-only statics = 72%; + mode-2 residual = 100%. Drop high modes **WITHOUT** residuals = lose statics. Drop **WITH** = engineering.

EX-Q: blade to 50 Hz (modes 8/22/47/130/260 Hz):

Keep 3 ($m = 3$, $n = \text{thousands}$). Modes 4-5 contribute stiffness-like leftovers -> residual term. Response error ~ 1 - 2% in-band. **[KEY]** This paragraph justifies every modal reduction in industry.

Laplace MDOF: global poles **PROVEN** by example:

$$[H(s)] = \text{adj}[B(s)] / \det[B(s)] \quad (B \text{ symmetric} \rightarrow H \text{ symmetric!})$$

FRF	Poles (global!)	Residues (local)
h_{11}	$-z_1 w_1 + -j w_1, -z_2 w_2 + -j w_2$	M1: 0.276, M2: 0.724
$h_{12} = h_{21}$	SAME poles	M1: +0.447, M2: -0.447
h_{22}	SAME poles	M1: 0.724, M2: 0.276

[KEY] POLES identical down the column (denominator shared); RESIDUES all different (numerators per i, j).

Global vs local - now a TABLE, not a slogan. **[LAB]** Fit giving different freqs per FRF? Your estimator is sick - poles are global, enforce it (global curvefitting, Ch.5).

-> Reciprocity sketch: B symmetric -> $\text{adj}(B)$ symmetric -> H symmetric -> $H_{ij} = H_{ji}$. Ch.1's row/column rule, derived from matrix symmetry in ONE line.

(2.3C) MDOF III: sums, beam, domains

FRF summation: read it like a story:

$$h_{ij}(\omega) = \sum_r A_{ijr} / (\omega - \omega_r) \quad [+ \text{conj}] \quad (\text{residue face})$$

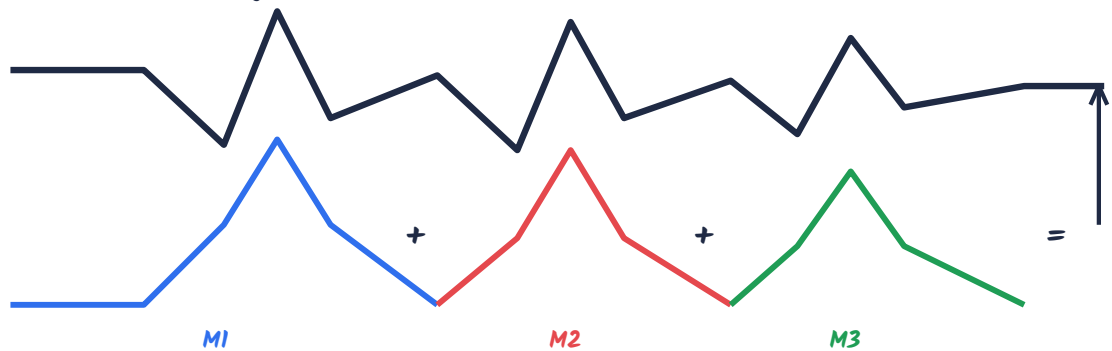
$$h_{ij}(\omega) = \sum_r u_{ir} u_{jr} / [m_r(\omega_r^2 - \omega^2 + j 2 \zeta_r \omega \omega_r)] \quad (\text{shape face})$$

my sketch: modal filter bank (input x shapes -> sum)



-> Input filter u_{jr} : shape at the HAMMER (push a node -> mode starves). SDOF amplifier: part 2.2, peaks at ω_r .
Output filter u_{ir} : shape at the SENSOR (sense a node -> mode hides).

my sketch: FRF = sum of modal SDOF humps
SUMMED FRF (what you measure)



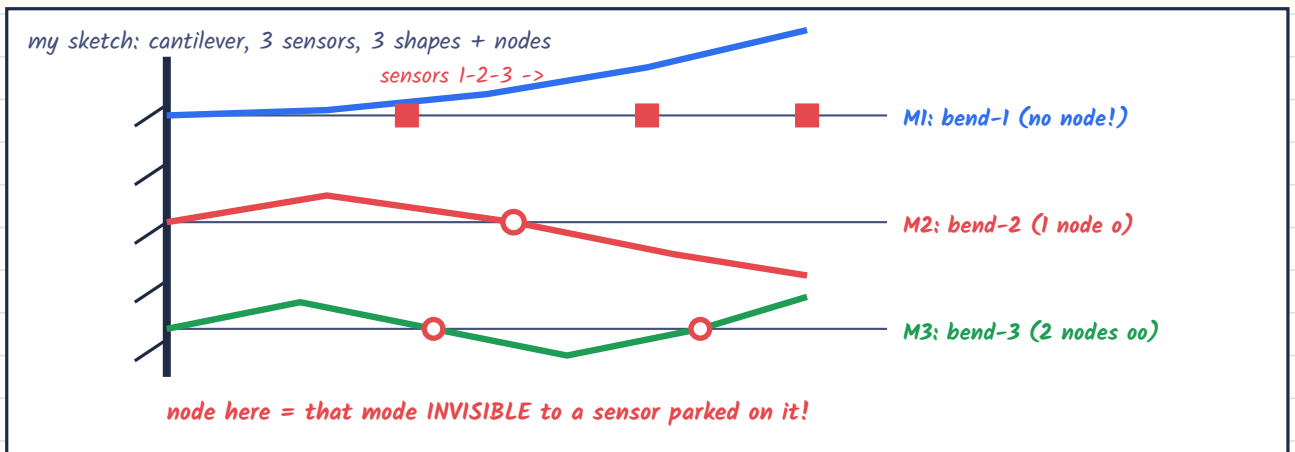
[KEY] The summed curve is what you MEASURE; the humps are what you EXTRACT. Curvefitting = unmixing paint back to primary colors.

Sine at any frequency: who answers? (book's 2DOF demo):

Excitation	Mode 1 says	Mode 2 says	Total looks like
Below both	Leads (soft, big)	Follows weakly	~ Mode 1
Near mode 2	Quasi-static leftover	RESONATES	~ Mode 2
Between	Fades with phase	Rises with phase	Hybrid (Ch.1 ODS!)
Above both	Mass-like, tiny	Mass-like, tiny	Small, mass-ruled

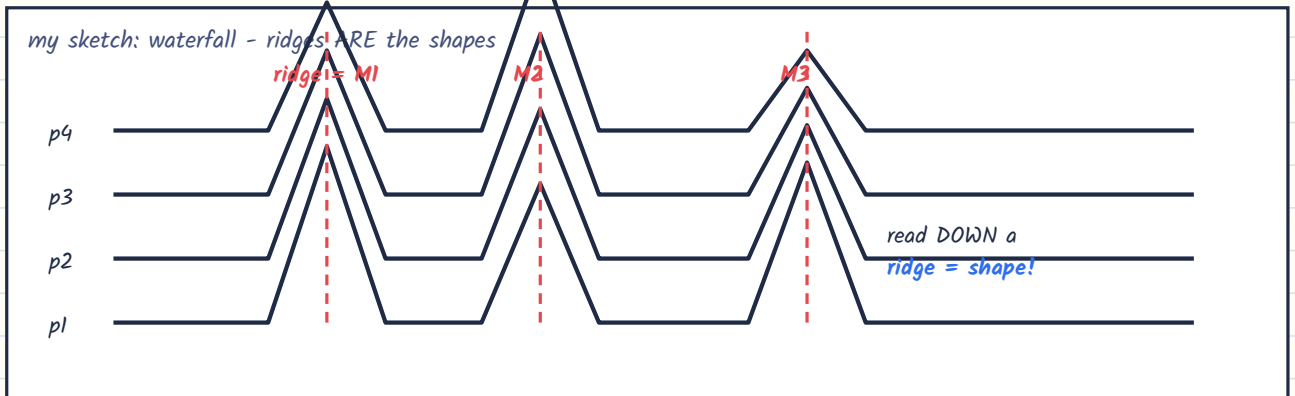
Time traces below each case (book Figs 2.35-37): single clean sine near a mode (one singer), beating/complex between (two singers arguing). **[LINK]** Frequency domain separates the singers; time domain mashes them - Ch.1's lesson, quantified.

Cantilever beam, 3 points: guided tour:



FRF	Role	Watch for
h33 (drive, tip)	Sum of oscillators; peak-valley alternation	Per-mode decomposition beneath
h32, h31 (cross)	Same poles, new residue weights	A mode shrinks as force nears its node
ROW (roving hammer)	Fixed accel reference	Ch.1 rule: DERIVED now
COLUMN (fixed shaker)	Fixed force reference	Read mode 1 down column 3!

Illustrative shapes (book-style): M1 = [0.25, 0.65, 1.0], M2 = [0.6, 1.0, -0.4] (node between 2-3!), M3 = [1.0, -0.5, 0.3]. **[KEY]** M2's TIP value is NEGATIVE: every mode-2 imag peak with tip involved flips DOWN. Sign = shape polarity, not error.



EX-R: read the waterfall:

Four FRFs (tip ref), M1 ridge heights [0.5, 1.2, 2.1, 3.0] -> normalize: [0.17, 0.4, 0.7, 1.0] = bending shape. M2 ridge: [0.9, 1.5, 0.2, -1.1] -> node between points 3-4! **[LAB]** Read DOWN a ridge = shape. Read ACROSS a curve = spectrum. Two readings, one plot.

Time-frequency-modal: three doors, one room:



Door	Math object	Convert with
Time	$x(t) = \text{SUM PHlr pr}(t)$: decaying sines	Laplace/FFT -> freq
Frequency	$H(j\omega) = \text{SUM SDOF humps}$	Inverse FFT -> time
Modal	$m_i \ddot{p} + c_i \dot{p} + k_i p = f_i + \text{shapes}$	[U] both ways

Turbine blade: the full computation (2.3.9):

-> Impulse near root (cantilever model), tip response wanted: eigensolve FEM -> modal $m/c/k$ + modal force per mode -> solve 3 trivial SDOF time responses -> PROJECT BACK with shapes -> add. Total tip trace with each mode's share beneath.

[KEY] This loop is EVERY commercial solver's inner core (Nastran/Ansys/Abaqus modal dynamics). The book's GIF animation shows the shares evolving - motion beats equations for gut feel.

[TRY] EX-S: blade modes 8/22/47 Hz ($z = 1/1.5/2\%$). Hit near root (all modal forces strong). Which mode rings LONGEST? 8 Hz (lowest $z \times w$: decay $1/(z w) = 2.0 \text{ s}$ vs 0.48/0.17 s). Tip trace after 1 s $\sim$ pure mode 1 - high modes already asleep!



(2.4) The whole EMA loop - block by block

my sketch: the whole EMA loop (Fig-2.55 spirit)

FORWARD: predict



BACKWARD: identify



Block	Does	Owens the error if wrong
FEM assume	Guesses M, K, BCs	Joints/BCs (Sofia's bracket!)
Eigensolve	Freqs + shapes, NO damping	Mesh too coarse, missed modes
Laplace H(s)	adj/det -> poles + residues	Non-proportional damping model
Synthesize FRF	Predict ANY hij	Truncation without residuals
Measure + FFT	Real F, real X	Leakage, mass loading, bad cal
Curvefit	Extract poles + residues	Wrong mode count, narrow band
Validate	Test vs FEM, update	Skipping it (the actual sin)

[KEY] Forward = predict from assumptions. Backward = identify from data. SAME objects (poles/residues) both ways. The gap = modeling + measurement error. Career = shrinking both. Next: Ch.3 signals, Ch.4 excitation, Ch.5 estimators.



(Deep 1) SDOF derivations - earn every formula

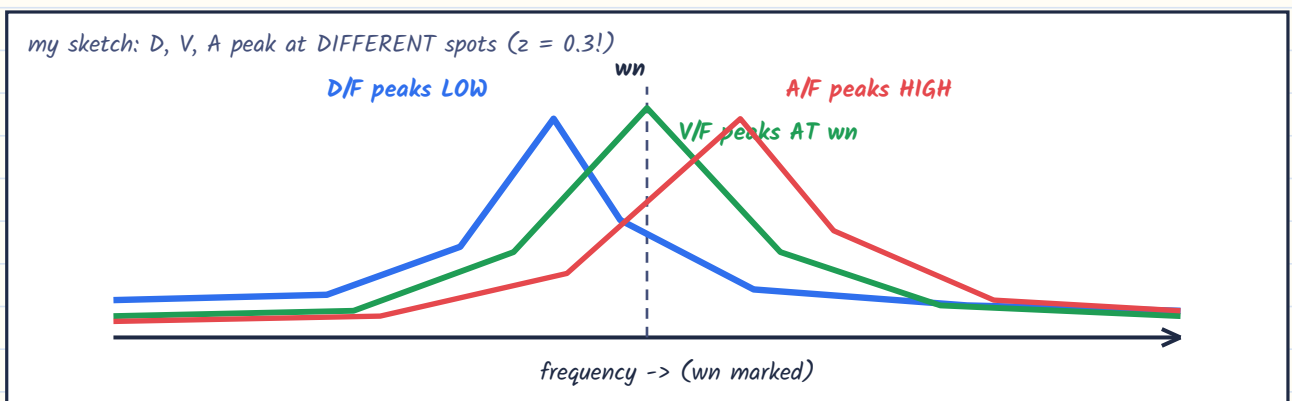
How wrong is $w_d \sim w_n$? (error table - quote with confidence):

z	1%	2%	5%	10%	20%	30%
w_d/w_n	0.99995	0.99980	0.99875	0.99499	0.97980	0.95394
Error of $\sim$	0.005%	0.02%	0.13%	0.5%	2.0%	4.6%

[KEY] Steel structures ($z \sim 1-3\%$): $w_d = w_n$ for all practical purposes. Rubber mounts ($z \sim 15\%+$): do the sqrt - 1%+ error bites in tuning.

DERIVE: where each peak REALLY sits ($d/dw = 0$):

- > D/F: minimize $(1-b^2)^2 + (2zb)^2$: derivative $4b(b^2 - 1 + 2z^2) = 0 \rightarrow b_D = \sqrt{1 - 2z^2}$ (needs $z < 0.707$ or NO peak!).
- > V/F: maximize $b^2/\text{denom} \rightarrow b_V = 1$ EXACTLY (velocity resonance = undamped fn - the purist's resonance).
- > A/F: maximize $b^4/\text{denom} \rightarrow b_A = 1/\sqrt{1 - 2z^2}$ (mirror of D above w_n).



z	5%: b_D / b_A	10%: b_D / b_A	30%: b_D / b_A
Peak spots	0.9975 / 1.0025	0.9899 / 1.0102	0.9055 / 1.1044

[TRAP] At $z = 30\%$ the D and A 'resonances' sit 20% APART. 'Resonance at 100 Hz' without naming the FRF form is meaningless past $z \sim 10\%$.

DERIVE: real/imag split (co-quad from algebra):

- > $H = 1/[(k - m\omega^2) + j(c\omega)]$: multiply top+bottom by conjugate:

$$\text{Re} = (k - m\omega^2)/D, \text{Im} = -(c\omega)/D, D = (k - m\omega^2)^2 + (c\omega)^2$$

- > At $\omega = \omega_n$: $k - m\omega^2 = 0 \rightarrow \text{Re} = 0, \text{Im} = -1/(c \omega_n)$ (pure imag peak!). Below: $\text{Re} > 0$ (stiffness-like). Above: $\text{Re} < 0$ (mass-like). The zero-crossing IS the resonance.

EX-T: full complex FRF at 3 spots ($m=2$, $k=8000$, $c=8$):

Static: $1/8000 = 1.25e-4$.

$w=30$: den $6200+j240 \rightarrow$ **Re $1.61e-4$, Im $-6.2e-6$** , $|I|$ $1.61e-4$, phase -2.2 deg (static-like, tiny lag).

$w=63.25$ (reso): den $j506 \rightarrow$ **Re 0 , Im $-1.98e-3$** , phase **-90 deg**, $15.8x$ static (= Q. of course).

$w=120$: den $-20800+j960 \rightarrow$ **Re $-4.80e-5$, Im $-2.2e-6$** , phase -177 deg (mass world, nearly flipped).

[KEY] Watch Re change SIGN across reso while Im stays negative and peaks: THE co-quad signature, computed not drawn.

[TRY] EX-U: velocity resonance is EXACTLY ω_n - verify: $|V| = \omega|H|$. $w=60$: $60 \times 1.072e-3 = 0.0643$. $w=63.25$: $63.25/506 = 0.125 = 1/c$. $w=66$: $66 \times 1.128e-3 = 0.0744$. Peak AT ω_n , value $1/c$: mobility resonance in one line.

SKETCH-PROOF: why Nyquist is a circle (mobility):

$\rightarrow V/F = j\omega/(k-m\omega^2+jc\omega)$. With a page of algebra it rearranges to $|V - 1/(2c)| = 1/(2c)$: points equidistant from $(1/2c, 0) =$ **CIRCLE diameter $1/c$** through origin + top.

$\rightarrow$ Resonance = TOP of circle (max imag, fastest sweep). Diameter reads damping DIRECTLY (big circle = light damping). Circle-fit = the grandfather of estimators.

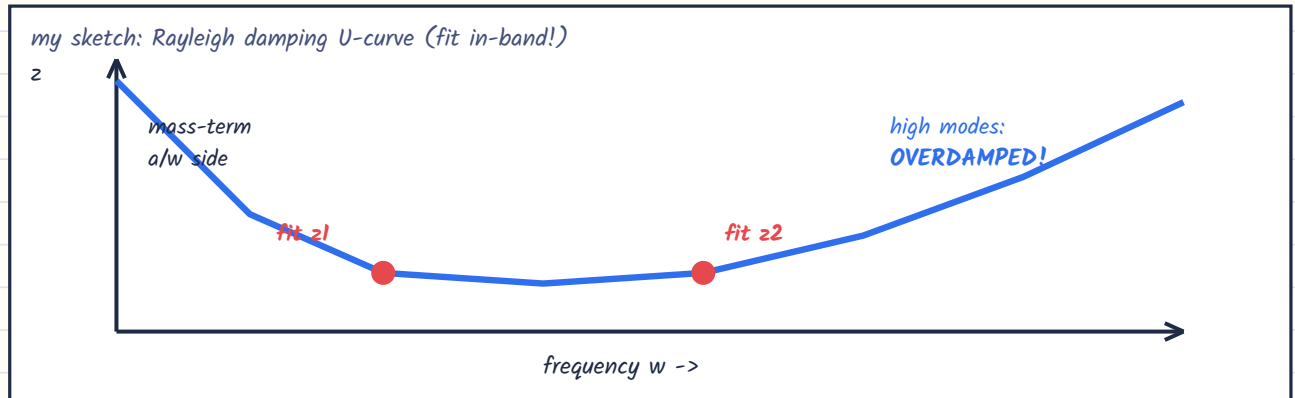
[LAB] Sweep-speed trick: equal-freq data points BUNCH at resonance (phase rushing) and spread elsewhere. Bunch location = resonance even before you fit. - Lena



(Deep 2) Damping lab + reading protocol

Rayleigh damping: fit z at TWO freqs, get C for free:

-> Assume $C = aM + bK$ -> modal: $c_i = a m_i + b k_i$ -> divide by $2 m_i \omega_i$: $z_r = a/(2\omega r) + b \omega r/2$ (mass-term falls, stiffness-term rises: U-curve!).



EX-V: fit $z1=2\%$ at $\omega1=6.18$, $z2=3\%$ at $\omega2=16.18$:

$a/12.36 + 3.09 b = 0.02$; $a/32.36 + 8.09 b = 0.03$. Solve: $b = 0.00324$, $a = 0.124$. Check mode 2: $0.124/32.36 + 0.00324 \times 8.09 = 0.0038 + 0.0262 = 0.030$. EXACT.

Extrapolate: $\omega=30 \rightarrow z = 5.1\%$; $\omega=60 \rightarrow 9.8\%$ [TRAP] Stiffness-term overdamps high modes. Rayleigh is a LOCAL fit: trust it INSIDE $[\omega1, \omega2]$, nowhere else. - Meera

Damping model	Force law	z_r vs ω	Use when
Viscous	$c \dot{x}$ (ellipse $\sim \omega$)	You set per mode	Default; Ch.2 world
Rayleigh	$aM + bK$	U-curve (fit 2 pts)	Time simulation needing C
Hysteretic	$k(1+j \eta)$ (ellipse const)	$\sim \eta/2$ flat	Rubber/joints; freq-domain only!

Hysteretic (structural) damping: stiffness carries a constant imag part η (loss factor, $\eta \sim 2z$ at reso). Loop area per cycle = $\pi \eta k X^2$ (amplitude-INDEPENDENT, unlike viscous $\pi c \omega X^2$). [TRAP] $H(j\omega) = 1/(k - m\omega^2 + j \eta k)$ has NO causal time twin - freq-domain ONLY. Time-sim it and physics breaks.

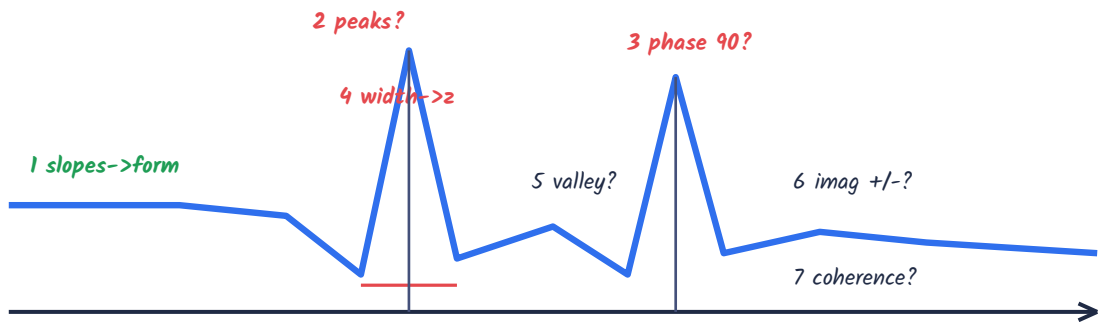
Shape scaling: 3 normalizations, same pattern:

Scaling	Mode-1 shape (EX-M)	Residue All,1	Use
Unscaled ($u1 = 1$)	$[1, 1.618]$, $m = 3.618$	$1 \times 1/3.618 = 0.276$	Sketching pattern
Max = 1	$[0.618, 1.0]$	Same 0.276 (scale cancels!)	Plots, slides
Mass-normalized	$[0.526, 0.851]$, $m = 1$	$0.526 \times 0.526 = 0.276$	RESPONSE math, residues

[KEY] Residues are scaling-INVARIANT: $(a u)(a u)/(a^2 m) = \text{same } A$. Shapes carry PATTERN; residues carry LEVEL. [TRY] EX-X: verify $A_{22,1}$ all three ways: $1.6182/3.618 = 12/1 = 0.8512 = 0.724$. Three doors, one number.

PROTOCOL: read any FRF in 60 seconds:

my sketch: read any FRF in 60 seconds (7 checkpoints)



#	Check	How	Fail means
1	Form?	Far slopes (0,-2)/(+1,-1)/(+2,0)	Mislabeled channels
2	Peak candidates?	Local maxima	Candidate only, not verdict!
3	Resonance?	Phase ~ 90 crossing	No 90 = no resonance
4	Damping?	Width $\rightarrow z \sim Dw/2w$	Too sharp? leakage suspects
5	Drive?	Valleys alternate?	Cross FRF or contaminated drive
6	Polarity?	Imag sign per mode	Sign flip = shape node crossed
7	Trust?	Coherence ~ 1 + repeat	Retest, do not fit garbage

Drive-point failure gallery (diagnose in seconds):

Case A - valley MISSING between peaks: neighbor mode flooding in, or overload flattening. Fix: finer resolution, check input levels.

Case B - imag peaks opposite sides on a 'drive' FRF: channels SWAPPED (force/response crossed) or wrong DOF. Fix: re-label, re-measure.

Case C - phase mushy, no clean flip: leakage / noise / nonlinearity. Fix: window, average, burst/chirp, lower level.

(Deep 3) MDOF extras + Ch.3 bridge

Rigid-body modes: the zeros that matter:

- > Free-free (no constraints): K SINGULAR -> 6 ZERO eigenvalues (3 translate + 3 rotate). Shapes = rigid motions (no strain, no stress). FRF low end = mass line $-1/(m \omega^2)$.
- > Test trick: hang the part on SOFT bungees (suspension $f < 10\%$ of first flex mode) -> rigid modes park near 0 Hz, flex modes untouched. Stiff support = free-free LOST.

[TRAP] Zero-frequency 'resonance' with flat shape = rigid mode, NOT a flex mode. Fitting it as elastic bends neighbors. - Viktor

Beyond proportional: complex modes + checks (honest boxes):

Real vs complex modes:

Proportional C -> shapes REAL (all DOFs in/180 deg together). General viscous C (real joints!) -> shapes COMPLEX (traveling-wave phase spread). **[KEY]** Ch.2 assumes proportional; Ch.5 estimators + Ch.9 cases handle the wild. If your 'shape' will not animate cleanly - complex-mode suspect.

Pseudo-orthogonality + MAC (preview with teeth):

Check test shapes $\{e\}$ against FEM: $\{e\}(T) [M_{fem}] \{e\}$ should be $\sim$ diagonal (pseudo-orthogonality). MAC = normalized shape correlation 0-1 (diagonal ~ 1 = match). Full MAC + validation = Ch.5 territory; the FORMULAS live here because Ch.2 owns orthogonality.



BRIDGE to Ch.3: damping DICTATES your measurement:

To resolve a peak you need SEVERAL spectral lines inside its half-power bandwidth: rule of thumb $Df \leq (2 z f_n)/5$ (5+ lines across $D\omega$). Light damping = LONG records ($T = 1/Df$) or leakage eats you.

$f_n = 100 \text{ Hz}, z$	Bandwidth $D\omega$	Need $Df \leq$	Record $T \geq$	Verdict
0.5%	1.0 Hz	0.2 Hz	5 s	Long + averaging!
1%	2.0 Hz	0.4 Hz	2.5 s	Careful windows
2%	4.0 Hz	0.8 Hz	1.25 s	Routine
5%	10 Hz	2 Hz	0.5 s	Easy peak

[KEY] THIS table is why Ch.2 comes before Ch.3: theory tells you what resolution to BUY. $z = 0.5\%$ with $Df = 1 \text{ Hz}$ = measuring a needle with a ruler.

EX-Y: residual bookkeeping (EX-M hll):

Exact static 0.0100. Mode-1-only: 0.00724. Residual $R = 0.00276$ (mode-2's quasi-static gift). In fitting: $h_{ij} \sim \text{SDOF1} + R$ (+ mass-residual at high f). **[LAB]** Always plot data-vs-synthesis OUTSIDE the fit band: drift = missing/broken residuals. - Viktor

EX-2: the wandering pole (diagnose!):

Fit says $f_1 = 50.1$ Hz from h11 but 51.3 from h22. Checklist: (1) band too tight around 50? (2) neighbor at ~55 hiding? (3) reference/level changed between runs? (4) temperature drift? (5) coherence dip at 51? **[KEY]**

Fix: widen band, include neighbor, GLOBAL refit. Poles are global - wandering poles indict the FIT, not physics.

Exam + lab traps (the wall of pain):

Trap	Truth
rad/s vs Hz in formulas	$\omega_n, \omega_d, D\omega$: rad/s. f_n, Df : Hz. Convert: $\omega = 2\pi f$. #1 exam killer.
$z = 2$ vs 2%	$z = 0.02$! Percent/decimal slips move poles 100x.
-3 dB from WHERE	From the PEAK, not full-scale. Wrong ref = fantasy z .
Log-dec on multi-mode	Beating fakes decay. Single-mode ONLY (or filter first).
kN/mm to N/m	$\times 1e6$! 30 N/mm = 30,000 N/m. Units kill FEMs daily.
Phase sign	Lags are negative (response AFTER force). Say 'lag 90', not '-90 or +270?'
Nyquist direction	Sweep passes resonance counter....: note YOUR analyzer's convention once, keep it.
Coherence = calibration?	NO: coherence 1.0 with wrong sensitivity = precisely wrong. Calibrate!

[LINK] Solver gotchas (from the chooser table): Lanczos can SKIP close modes (shift + verify count!); rigid modes come out $\sim 1e-3$ Hz not 0 (numerical dust - fine); always compare mode COUNT vs expected in-band.



(Deep 4) Zeros, synthesis, transients

Zeros: where drive-point FRFs touch ZERO (derived!):

-> Undamped $h_{11} = A_1/(w_{12}^2 - w^2) + A_2/(w_{22}^2 - w^2)$. Set = 0: $A_1(w_{22}^2 - w^2) + A_2(w_{12}^2 - w^2) = 0 \rightarrow w^2 = (A_1 w_{22}^2 + A_2 w_{12}^2)/(A_1 + A_2)$.

EX-AA: the anti-resonance of EX-M (compute it!):

$A_1 = 0.276$, $A_2 = 0.724$, $w_{12} = 38.2$, $w_{22} = 261.8$: $w^2 = (0.276 \times 261.8 + 0.724 \times 38.2)/1.0 = (72.36 + 27.64) = 100.0 \rightarrow w = 10.0 \text{ rad/s (1.59 Hz)}$.

[KEY] Sits BETWEEN $w_1 = 6.18$ and $w_2 = 16.18$: poles and zeros INTERLACE (drive points only!). Valley depth -> infinity undamped; damping fills it in. You just PREDICTED a valley from residues.

Physics of zeros: at w_z the two modal contributions are EQUAL and OPPOSITE - the drive point stands STILL while the rest moves (vibration absorber principle!). **[TRAP]** Cross FRFs: zeros wander anywhere - no interlacing, no alternation. Alternation is a DRIVE-POINT privilege.

EX-AB: synthesize h_{12} AT mode-2 resonance (phase flip!):

-> $w = 16.18$, $z_1 = 2\%$, $z_2 = 3\%$. Mode 1: $0.4472/(-223.6 + j4.0) \sim -0.0020$ (real leftover). Mode 2: $-0.4472/(j15.71) = +j0.0285$ (NEGATIVE residue!).

$h_{12}(16.18) \sim -0.0020 + j0.0285$ (1.1 0.0286, angle +94 deg)

[KEY] Usual resonance: -90 deg. NEGATIVE residue: +94 deg - phase FLIPPED 180. Cross-FRF phases read residue SIGNS: your phase plot is a shape-polarity map.

Transient trio: impulse, step, release (formulas + feel):

Input	Response $x(t)$ (underdamped)	Signature
Initial x_0 (release)	$x_0 e^{-(z\omega t)} [\cos \omega t + z/\sqrt{1-z^2} \sin \omega t]$	Ring-down (log-dec lives here)
Impulse I	$(I/m\omega d) e^{-(z\omega t)} \sin \omega t$	Pure sine envelope: THE impulse response
Step F_0	$(F_0/k)[1 - e^{-(z\omega t)}(\cos \omega t + \dots)]$	Overshoot $\sim e^{-(\pi z/\sqrt{1-z^2})}$

EX-AC: release EX3 rig from 5 mm ($z = 3.16\%$):

$T_d = 2\pi/63.24 = 0.0993 \text{ s}$. Half-period decay: $e^{-(0.0316 \times 63.25 \times 0.0497)} = e^{-0.0993} = 0.905$. First negative peak $\sim -4.5 \text{ mm}$ at 0.0497 s. Each swing keeps $\sim 90\%$: after 10 cycles $\sim 35\%$ left. **[TRY]** Count swings to half amplitude: $\ln 2/(2\pi z) = 3.5$ cycles. Quick lab estimate of z from ANY ring-down!

Beats: why log-dec dies on close modes:

$x = \sin(2\pi 10 t) + \sin(2\pi 10.5 t) = 2 \cos(2\pi 0.25 t) \sin(2\pi 10.25 t)$: 10.25 Hz carrier inside a 2-second breathing envelope. Peaks wobble from INTERFERENCE, not damping. **[TRAP]** Log-dec on beats = random numbers. Frequency domain separates the singers; time domain records their argument.

(Deep S) State-space, MIMO, units, links

State-space: first-order form (where estimators live):

-> Stack $y = [x; v]$: then $y' = A y + B f$ with $A = \begin{bmatrix} [0, I], [-\text{inv}(M)K, -\text{inv}(M)C] \end{bmatrix}$. n 2nd-order eqns -> $2n$ 1st-order.

-> Eigenvalues of A = THE $2n$ POLES (complex pairs, always). Eigenvectors = complex modes + conjugates. Real-modes world (Ch.2) = special symmetric case of this general machine.

[LINK] Why care NOW: every modern estimator (Ch.5: LSCE, ERA, PolyMAX...) fits THIS form, then translates back to freqs/shapes. Ch.2 is the translation dictionary.

DERIVE: proportional damping diagonalizes (3 lines):

-> Given $C = aM + bK$: $U(T) C U = a U(T)M U + b U(T)K U = a \text{diag}(m) + b \text{diag}(k) = \text{diag}(c)$. DONE - orthogonality did it again.

$$c_i = a m_i + b k_i, z_i = c_i / (2 m_i \omega_i) = a / (2 \omega_i) + b \omega_i / 2$$

[TRY] EX-AD: EX-M with $z_1 = 2\%$, $z_2 = 3\%$: $c_1 = 2 \times 0.02 \times 6.18 \times 3.618 = 0.894$; $c_2 = 2 \times 0.03 \times 16.18 \times 1.382 = 1.342$ (modal units!). Test gives z ; models eat c - this line converts.

Test alphabet: SISO to MIMO (what you buy per setup):

Setup	Inputs x Outputs	You get	Kills
SISO	1 hammer + 1 accel	ONE hij (spot checks)	Quick looks, tiny rigs
SIMO	1 fixed shaker x roving accels	ONE COLUMN	Single-ref shapes
MISO	Roving hammer x 1 accel	ONE ROW	Fast field surveys
MIMO	Many shakers x many accels	MANY rows/cols	Close modes, big iron

[KEY] MIMO costs rigs + channels but SEPARATES close/coupled modes single-ref tests fuse. Ch.4 picks; Ch.2 explains WHY (independent columns = independent equations).

Units cheat sheet (SI modal survival):

Qty	Unit	Trap
Force	N	Hammer tips rated in N - respect them
Disp / Vel / Acc	m / m-s / m-s ² (or mm, g!)	mm vs m = 1000x; g = 9.81
Stiffness	N/m	N/mm x 1000! kN/mm x 1e6!
Freq	Hz (f) vs rad/s (w)	$w = 2 \pi f$. Formulas want w.
FRF mag	m/N, (m/s)/N, (m/s ²)/N	Compare ONLY same form!
dB	$20 \log_{10}(H /\text{ref})$	No ref stated = graffiti, not data

Ch.2 results -> where they get USED (dependency map):

Ch.2 result	Used in	As
Poles (global)	Ch.5 estimators	Fit targets (stabilization!)

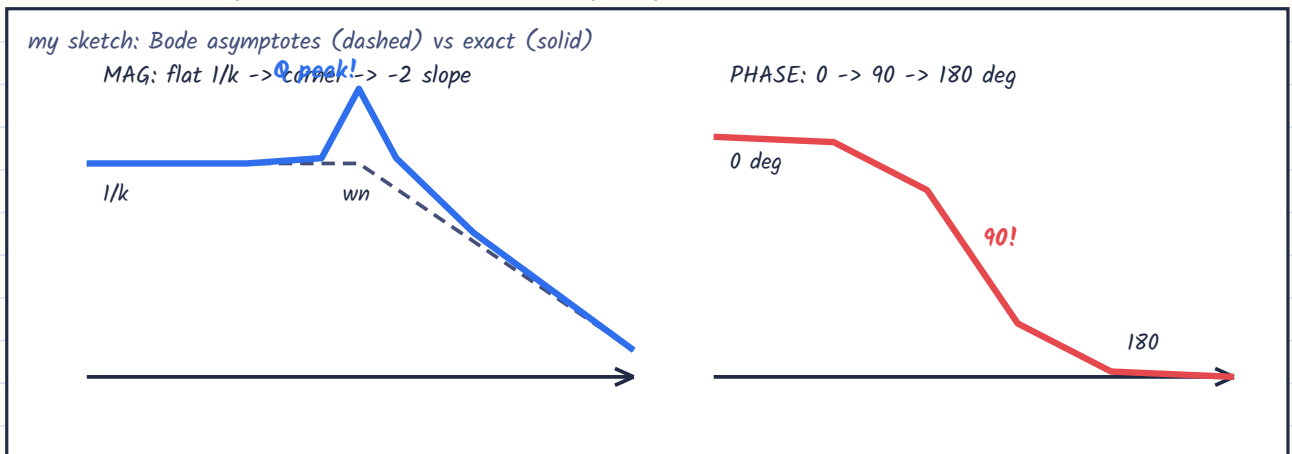
Ch.2 result	Used in	As
Residues/shapes	Ch.5 + animation	Shape extraction + display
Bandwidth Dw	Ch.3 acquisition	Resolution + record length spec
Reciprocity	Ch.4 MIMO checks	hij-vs-hji quality gate
Orthogonality	Ch.5 validation	Pseudo-orth + MAC vs FEM
Modal superposition	Every solver	Transient/forced response core
D/V/A slopes	Every test day	First-plot channel detective

Practice round 2 (P9-P16) + answers:

- * **P9.** EX-M h11 zero: recompute blind, then state the interlacing rule. [A: $\omega z = 10.0$ rad/s; drive zeros sit BETWEEN poles.]
- * **P10.** Fit Rayleigh through $z = 1.5\%$ at 5 Hz and 2.5% at 20 Hz. Give a, b. [A: $\omega = 31.4/125.7$; $b = 2.86e-4$, $a = 0.795$. Check both!]
- * **P11.** h12 at mode-2 reso gave $+94$ deg. Explain the sign to a fresher. [A: Negative residue (shapes oppose) rotates phase 180 from the usual -90 .]
- * **P12.** Ring-down halves in 3.5 swings. z ? [A: $\ln 2 / (n \cdot 2 \pi)$ with $n = 3.5$: $z \sim 3.15\%$.]
- * **P13.** 10 + 10.4 Hz equal sines: beat period? [A: Envelope $|\cos|$ at 0.2 Hz: 5 s breathing. Log-dec = garbage here.]
- * **P14.** Stiffness 45 N/mm in N/m? Static sag of 200 kg on it? [A: 45,000 N/m; sag = $1962/45000 = 43.6$ mm.]
- * **P15.** V/F peak reads 60.0 Hz, A/F peak 60.5 Hz. z estimate? [A: $b_A = 1/\sqrt{1-2z^2} = 60.5/60 \rightarrow z \sim 9.1\%$. (Heavy! rubber mount?)]
- * **P16.** MIMO vs SIMO for a symmetric plate with twin modes? [A: MIMO – single ref fuses repeated roots; independent columns split them.]

(Deep 6) Bode, transients, isolation

Bode sketching masterclass (D/F, log-log):



Piece	Rule	EX3 rig values
Low-f asymptote	Flat at $1/k$	$1/8000 = 1.25e-4$
Corner	At ω_n (10.07 Hz here)	63.25 rad/s
High-f asymptote	-2 slope (-40 dB/dec)	Mass line $1/(m \omega^2)$
Peak (light z)	$Q = 1/(2z)$ above corner crossing	$15.8x \rightarrow 1.98e-3$
Phase	0 far low, 90 AT corner, 180 far high	~ 1 decade each side

[TRY] EX-AE: sketch EX3 D/F from asymptotes ONLY, then check 3 exact points: $\omega=30$ ($1.61e-4$ vs flat-ish $1.3e-4$: close!), $\omega=63.25$ (peak, asymptotes useless - that IS the lesson), $\omega=120$ ($4.8e-5$ vs mass-line $1/(2 \times 14400) = 3.5e-5$: same street). Asymptotes navigate; exact math docks.

[TRAP] Bode peak error: asymptotes MISS the peak by design - that gap IS Q . Never read damping off asymptotes; read it off the exact peak/hp width.

Complex pair = ONE real sine (why pairs matter):

$\rightarrow x = A e^{(p_1 t)} + A^* e^{(p_2 t)}$, $p_{1,2} = -z \omega_n \pm j \omega_d$. Pair them: $x = 2|A| e^{(-z \omega_n t)} \cos(\omega_d t + \text{angle})$.
Imaginary parts CANCEL - reality restored!

[KEY] This is WHY poles come in pairs: lone complex pole = complex (impossible) response. Conjugates are reality's bodyguards.

Transient + steady-state: how long to WAIT (lab gold):

$\rightarrow$ Start a sine: $x = \text{transient (decays } e^{(-t/\tau)}, \tau = 1/(z \omega_n)) + \text{steady sine}$. Measure during transient = polluted FRF. Rule: wait **4-5 tau** (2% left).

EX-AF: EX3 rig ($z = 3.16\%$, $\omega_n = 63.25$):

$\tau = 1/(0.0316 \times 63.25) = 0.50$ s. Wait 2-2.5 s before trusting sine-dwell data. **[LAB]** Burst-random record length rule (Ch.4 preview): capture 4-5 tau of DECAY inside one record $\rightarrow$ leakage-free WITHOUT windows. Theory just sized your acquisition! - Lena

z at 10 Hz	τ	Wait 4 τ	Lesson
0.5%	3.2 s	13 s (!!)	Bells need patience
2%	0.8 s	3.2 s	Typical steel
10%	0.16 s	0.6 s	Rubber: quick + easy

COUNCIL EXTENSION: transmissibility (isolation design):

-> Force to ground (or ground to mass): $T = \sqrt{1 + (2zb)^2} / \sqrt{(1 - b^2)^2 + (2zb)^2}$. Isolate ($T < 1$) needs $b > \sqrt{2}$ - mounts tuned BELOW 0.7x the disturbance!

EX-AG: isolate a 25 Hz machine (want 88% cut):

Pick mounts $f_n = 8$ Hz ($b = 3.125$), $z = 0.05$: $T = \sqrt{1.0977} / \sqrt{76.93} = 1.048 / 8.77 = 0.119$ -> 88% isolated. **[TRAP]** More damping WORSENS high-f isolation (numerator grows)! Soft + light beats stiff + dead. Designers get this backwards yearly.

[LINK] Dwell link (Ch.1 plate!): sine-dwell AT reso = steady-state sine after transients die: response = $|H| F$ at 90 deg lag. Ch.1's experiment was Ch.2's forced-response page, performed live.



(Deep 7) Quotient, absorber, stability, plans

Rayleigh quotient: guess a shape, bound w/ (from ABOVE!):

$$w^2(u) = [u(T) K u] / [u(T) M u] \geq w^2 \text{ (ANY } u\text{!)}$$

-> Any guessed shape gives freq ABOVE true mode-1 (energy truth: wrong shapes strain extra). Better guess -> tighter bound. Pre-computer engineers lived by this.

EX-AH: bound EX-M mode 1 (true w/2 = 38.2):

Guess [1, 1]: $Ku = [100, 0] \rightarrow u(T)Ku = 100$; $u(T)Mu = 2 \rightarrow w^2 = 50.0$ (bound, 31% high).

Guess [1, 1.5]: $Ku = [50, 50] \rightarrow 125$; mass 3.25 $\rightarrow w^2 = 38.46$ (0.7% high!). **[KEY]** Close guess $\rightarrow$ near-exact freq. Quotient turns sketches into numbers.

Zeros at work: the tuned absorber (design it!):

-> Attach small (m_a, k_a) tuned to the trouble freq: drive-point zero lands ON it $\rightarrow$ main mass stands STILL (its energy parked in the absorber). Skyscrapers + crankshafts run on this.

EX-AI: calm a 100 kg machine shaking at 25 Hz:

Tune absorber $w_{n,a} = 25$ Hz: $(2 \pi 25)^2 = 24674$. Pick $m_a = 5$ kg $\rightarrow k_a = 5 \times 24674 = 123 \text{ kN/m}$. Result: zero at 25 Hz on the machine (two NEW poles straddle it - check they are harmless!). **[TRAP]** Absorbers SPLIT one peak into two: retune the band, or trade one problem for two.

Zero physics you can FEEL: tuning-fork handle nearly still while tines scream (drive near a zero!); loudspeaker cabinet dead-spots; the 'node' of Ch.1 = zero of that FRF. Zeros are not valleys on paper - they are STILLNESS in metal.

Stability: poles must live LEFT (negative real!):

Pole home	Time says	Meaning
Left half (real < 0)	Decays	Stable: all Ch.2 systems
j axis (real = 0)	Rings forever	Undamped ideal / flutter edge
Right half (real > 0)	GROWS ($e^+ + t$)	UNSTABLE: flutter, chatter, squeal!

[KEY] Brake squeal, wing flutter, tool chatter = poles DEFECTED right (friction/aero feeding energy). Modal test + stability map = diagnosis. Ch.2's S-plane is also a stability radar.

Read animated shapes like a pro:

-> 1. Find NODES first (still lines) - they count the mode order. 2. Check phase colors: two-color flip-flop = real mode; rainbow crawl = complex/close modes. 3. Amplitude scale LIES (auto-scaled!) - compare within one animation only. 4. Slow the loop: twisting vs bending hides at speed.

[TRAP] Peak-picking error audit: neighbor leakage bends BOTH height and apparent freq. Close modes (Dw overlap) + strong neighbors = fit, do not pick. EX-M was kind (4x apart); real life often is not.

Study plans (pick your battle):



<i>Plan</i>	<i>Do</i>	<i>Skip (for now)</i>
1-day sprint	Map + KEY boxes + sketches + Banks 1-2 + revision	Derivations, masterclass arithmetic
1-week steel	ALL EX by hand + Deep 1-3 + Problems P1-P16	Deep 4-7 (second pass)
Interview armor	FAQ + rapid-fire + protocol + traps + EX-M story	Hysteretic/state-space detail
Lab ready	Protocol + failure gallery + resolution table + checklist	Proofs (keep as backup)

[LINK] Where Ch.2 NEVER goes (honest borders): nonlinear systems, rotating machinery (gyroscopics), acoustics, control-structure interaction. It conquers linear stationary vibration – which is 90% of paid modal work.



(Deep 8) Core proofs, fully shown

PROOF: orthogonality (the engine room):

- > Modes r, s satisfy $(K - \omega_r^2 M) u_r = 0$ and $(K - \omega_s^2 M) u_s = 0$. Premultiply mode- r eq by u_s^T : $u_s^T K u_r = \omega_r^2 u_s^T M u_r$.
- > Premultiply mode- s eq by u_r^T , transpose (K, M symmetric): $u_s^T K u_r = \omega_s^2 u_s^T M u_r$. SUBTRACT: $0 = (\omega_r^2 - \omega_s^2) u_s^T M u_r$.

$$\omega_r \neq \omega_s \Rightarrow u_s^T M u_r = 0 \text{ (and } u_s^T K u_r = 0)$$

[KEY] Distinct freqs -> automatic M/K-orthogonality. Repeated roots: orthogonality must be ENFORCED (Gram-Schmidt) - one more reason twins are trouble.

PROOF: modal FRF sum from the matrix equation:

- > Start: $(-\omega^2 M + j\omega C + K) X = F$. Expand $X = \sum q$ (shape combo!), premultiply by u_r^T : orthogonality kills ALL terms except r :

$$(-\omega^2 m_r + j\omega c_r + k_r) q_r = u_r^T F \text{ (ONE scalar eqn!)}$$

- > Solve q_r , substitute back $X = \sum u_r q_r$:

$$H(\omega) = \sum_r u_r u_r^T / (k_r - \omega^2 m_r + j\omega c_r)$$

[KEY] The whole chapter in 4 lines: expand, project, solve scalars, recombine. Residues ARE $u_r u_r^T / m_r$. Memorize THIS derivation, not the formula.

PROOF: log-dec formula (peaks one period apart):

- > $x(t) \sim e^{(-z \omega_n t)} \cos(\dots)$: peaks at $t, t+T_d$: ratio $x_1/x_2 = e^{(z \omega_n T_d)}$, $T_d = 2\pi/\omega_d = 2\pi/(\omega_n \sqrt{1-z^2})$. Take $\ln$:

$$d = \ln(x_1/x_2) = 2\pi z / \sqrt{1-z^2} \approx 2\pi z$$

- > Over n cycles: $d = (1/n) \ln(x_1/x_{n+1})$ (averages noise). Invert: $z = d/\sqrt{4\pi^2 + d^2}$.

PROOF SKETCH: $Q = 1/(2z)$ from half-power:

- > $|H|^2$ peak = $1/(2zb)^2 \dots$: half-power where denom doubles: $(1-b^2)^2 + (2zb)^2 = 2(2z)^2$ (light $z, b \sim 1$). Solve: $b^2 \sim 1 \pm 2z \rightarrow b \sim 2z \rightarrow \omega_d = 2z \omega_n$. $Q = \omega_n/\omega_d = 1/(2z)$.



(Deep 9) EX-M capstone + full index

Pole-zero map of EX-M (every number placed):

- > Poles ($z_1 = 2\%$, $z_2 = 3\%$): $p_1 = -0.124 \pm j6.179$, $p_2 = -0.485 \pm j16.173$ (left-half, stable, ringing).
- > Zeros of h_{11} (undamped): $\pm j10.0$ (ON the axis: perfect stillness possible). h_{12} : NO finite zeros (its numerator never vanishes - cross privilege lost!).

FRF	Mode 1 residue -> phase	Mode 2 residue -> phase
h_{11} (drive)	+0.276 -> -90 deg	+0.724 -> -90 deg
$h_{12} = h_{21}$ (cross)	+0.447 -> -90 deg	-0.447 -> +90 deg (FLIP!)
h_{22} (drive)	+0.724 -> -90 deg	+0.276 -> -90 deg

[KEY] This 3x3 IS the whole rig: poles say WHEN/DECAY, residues say WHERE/HOW MUCH, signs say WHICH WAY. Everything else is commentary.

Cross-check clinic: FIVE routes, one truth (all static):

(1) Invert K: $100/10000 = 0.0100$. (2) Residue sum: $0.00724 + 0.00276 = 0.0100$. (3) Zero-formula check: $(A_1 w_{22} + A_2 w_{12}) = 100.0 = w_{z2} \times 1.0$: consistent. (4) Low-f Bode asymptote -> flexibility line. (5) Quotient bounds w_1 from above. **[KEY]** When 5 independent paths agree, you do not HAVE the answer - you OWN it.

Every worked example in this book (34 total):

EX	Topic	Lives in
A	Quarter-car suspension (real numbers)	2.2A
B	Pole drill: three fates	2.2A
C	Sketch S-planes	2.2A
D	Amplification table	2.2A
E	Phase arithmetic	2.2A
F	Half-power z from 100 Hz peak	2.2B
G	Ring-down table	2.2B
H	z -Q-d conversions	2.2B
I	Same rig, off-resonance FRF	2.2B
J	Residue from modal mass	2.2B
K	Nyquist top + hp points	2.2B
L	Slope detective quiz	2.2B
M0	D/V/A conversion at 50 Hz	2.2B
M	2DOF MASTERCLASS (full FRF)	2.3A
N	2DOF intuition in words	2.3B



EX	Topic	Lives in
P	Modal forces $F=[1,0]$	2.3B
Q	Blade modes to 50 Hz	2.3C
R	Read the waterfall	2.3C
S	Blade hit: which modes sing	2.3C
T	Complex FRF at 3 spots	Deep-1
U	V-resonance exactly ω_n	Deep-1
V	Rayleigh fit (a, b)	Deep-2
X	Scaling invariance of A	Deep-2
Y	Residual bookkeeping	Deep-3
Z	Wandering-pole diagnosis	Deep-3
AA	Anti-resonance $\omega_z = 10$	Deep-4
AB	Synthesis + phase flip	Deep-4
AC	5 mm release decay	Deep-4
AD	Modal c from test z	Deep-5
AE	Bode asymptotes vs exact	Deep-6
AF	Wait-time from tau	Deep-6
AG	Isolation design (88%)	Deep-6
AH	Rayleigh quotient bound	Deep-7
AI	Absorber design	Deep-7

(Letters O and W unused: O confuses with 0, W with w. Notation hygiene!)

FAQ round 2 (deep cuts):

Q1. Why do drive zeros interlace but cross zeros wander?

A: Drive numerator is constrained by collocation (force+response same DOF): signs alternate. Cross numerators are free. - shared

Q2. Can residues be complex?

A: With proportional damping: real. General viscous: complex (phase spread). Real A = real modes. - shared

Q3. Why does Rayleigh overdamp high modes?

A: $b \omega/2$ term grows unbounded. Fit band edges are trust edges. - Meera

Q4. Bode vs Nyquist vs FRF: which do I plot first?

A: Bode mag (slopes+peaks), then phase (confirm), Nyquist only when fitting circles. - Lena

Q5. Absorber vs isolation mount: opposite?

A: YES: mounts push reso DOWN (avoid); absorbers plant a ZERO on the trouble. - shared

Q6. One-line Ch.2?

A: Poles say when, residues say where, shapes say how - everything else is bookkeeping. - council

Notation decoder (never confuse again):

Symbol	Means	Symbol	Means
w / W	rad/s freq / matrix	z	DAMPING RATIO (star!)
u / U	shape vector / modal matrix	q	modal coordinate
m / M	modal mass / mass matrix	H / h	FRF matrix / one FRF
A	residue (modal strength)	p / l	pole / eigenvalue (w^2)



(Bank 1) Formula bank - tear-out pages

SDOF core:

$$m \ddot{x} + c \dot{x} + k x = f(t)$$

$$\omega_n = \sqrt{k/m}, \quad \zeta = c / (2 m \omega_n), \quad z = c / c_c$$

$$s = -\zeta \omega_n \pm j \omega_d, \quad \omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$X/X_{st} = 1/\sqrt{(1-b^2)^2 + (2\zeta b)^2}, \quad b = \omega/\omega_n$$

$$\text{phase} = \text{atan2}(2\zeta b, 1-b^2): 0 \rightarrow 90 \rightarrow 180 \text{ deg}$$

$$z \sim D\omega/(2\omega_n), \quad Q = 1/(2\zeta), \quad d = \ln(x_1/x_2) \sim 2\pi z$$

$$H(s) = 1/(ms^2 + cs + k) = A_1/(s-p_1) + A_2/(s-p_2)$$

$$A_1 = 1/(j 2 m \omega_d) \text{ (cover-up)}$$

$$V/F = j\omega(D/F), \quad A/F = -\omega^2(D/F)$$

MDOF core:

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F\}$$

$$\det([K] - L[M]) = 0 \rightarrow Lr = \omega_r^2 \text{ (eigenvalues)}$$

$$([K] - \omega_r^2 [M])\{u_r\} = 0 \text{ (eigenvectors/shapes)}$$

$$u_i(T)[M]u_j = m_i d_{ij}, \quad u_i(T)[K]u_j = k_i d_{ij}$$

$$\{x\} = [U]\{p\}: m_i \ddot{p}_i + c_i \dot{p}_i + k_i p_i = f_i = u_i(T)\{F\}$$

$$[H(s)] = \text{adj}[B(s)]/\det[B(s)]: \text{poles GLOBAL}$$

$$h_{ij} = \text{SUM } u_{ir} u_{jr} / [m_r(\omega_r^2 - \omega^2 + j 2\zeta_r \omega_r \omega)]$$

$$A_{ijr} = u_{ir} u_{jr} / m_r \text{ (mass-normalized: } u_{ir} u_{jr})$$

$$H_{ij} = H_{ji} \text{ (symmetry } \rightarrow \text{ reciprocity)}$$

Symbols (nomenclature):

Symbol	Name	Unit / note
m, c, k / M, C, K	Mass, damping, stiffness (scalar/matrix)	kg, Ns/m, N/m
ω_n , ω_d , f_n	Natural / damped freq	rad/s, rad/s, Hz ($f_n = \omega_n/2\pi$)
z, ζ , Q, d	Ratio, critical c, quality, log-dec	-, Ns/m, -, -
s, p_1/p_2 , L	Laplace var, poles, eigenvalue	1/s; $L = \omega^2$
A, A_{ijr}	Residue (SDOF / MDOF pair)	Carries shape product
U, u, p, f	Modal matrix, shape, modal resp/force	$x = [U]p$

Symbol	Name	Unit / note
$H(s)$, $H(j\omega)$, h_{ij}	Transfer, FRF, i-j term	Response/input
b	Frequency ratio ω/ω_n	Tuning knob of Ch.2

(Bank 2) Misconceptions FAQ - the council's hit list

Q1. Is the peak exactly at f_n ?

NO (mostly). D/F peaks slightly BELOW ω_n , V/F EXACTLY at ω_n , A/F ABOVE. Light z : all $\sim \omega_n$. Say WHICH FRF!

Q2. Does damping lower the natural frequency?

ω_n NEVER moves. $\omega_d = \omega_n \sqrt{1-z^2}$ dips slightly. $z=0.3 \rightarrow \omega_d = 0.95 \omega_n$.

Q3. Negative imag peak = broken data?

NO - residue SIGN = shape polarity (EX-M mode-2 cross terms). All-same-side rule applies ONLY to drive points!

Q4. ODS = mode shape?

Only when exciting AT an isolated resonance (then ODS $\sim$ that mode). Elsewhere: a MIX (Ch.1 1.8, now with equations).

Q5. Can poles differ between FRFs?

True poles: NEVER (global). Fitted poles wandering = estimator/band trouble - enforce global fits (Ch.5).

Q6. More modes always better?

NO - wrong count bends neighbors. Right count + residuals beats mode-soup. (Viktor: 'fitting 6 modes into 3 peaks is astrology'.)

Q7. Is 90-deg phase enough to claim resonance?

Nearly - plus peak + coherence + repeatability. Phase is the lie detector, not the whole court.

Q8. Why do shapes need scaling?

Pattern vs LEVEL: [1,1.618] draws right; mass-normalized [0.526,0.851] RESPONDS right. Residues carry scale.

Q9. Zeros = anti-resonances?

At DRIVE points, FRF zeros sit between poles = valleys. Cross FRFs: zeros wander (no alternation rule).

Q10. Complex modes - when?

Non-proportional damping (real joints!) breaks real-mode math. Ch.2 assumes proportional; Ch.5+ handles the wild.

(Bank 3) Interview rapid-fire (12)

- * 11. Poles? Roots of char. eq: damping (real) + freq (imag), conjugate pairs.
- * 12. Residues? Pole strengths; MDOF: shape-i x shape-j. With poles: full DNA.
- * 13. Why one row/column? H symmetric (M/C/K symmetric) $\rightarrow H_{ij} = H_{ji}$.
- * 14. Node reference? $u_{jr} = 0$ zeroes the mode in the whole row/column.
- * 15. Half-power? $z = D\omega/(2\omega_n)$. Quick, single-mode-only, pros fit instead.

- * **16.** 90 degrees? Force-response lag AT resonance; most robust detector.
- * **17.** D vs V vs A? $\times$ jw per step; slopes (0,-2)/(+1,-1)/(+2,0).
- * **18.** Global vs local? Poles shared (denominator); residues per pair.
- * **19.** Orthogonality? Shapes decouple M,K $\rightarrow$ modal masses/stiffnesses.
- * **110.** $m \ll n$? Keep in-band modes; fold rest into residuals. Industry runs on this.
- * **111.** EMA vs FEM? Test = truth + damping; FEM = what-ifs. Loop them.
- * **112.** Nyquist potato? Overlap, noise/leakage, or nonlinearity. Investigate!

(Bank 4) Practice problems (8) + answers

- * **P1.** $m=4$, $k=9000$, $c=24$. Find f_n , z , poles, Q . [A: $\omega_n=47.4$, $f_n=7.55\text{Hz}$, $\zeta=379$, $z=6.3\%$, poles $-3+j47.3$, $Q \sim 7.9$]
- * **P2.** Peak 200 Hz, hp at 198.5/201.5. z ? Q ? Light/heavy? [A: $z=0.75\%$, $Q=67$, light - mind leakage!]
- * **P3.** Ring-down peaks 5.0, 4.1, 3.36. z ? [A: $d \sim \ln(5/4.1)=0.198$ avg $\rightarrow z \sim 3.15\%$]
- * **P4.** D/F at 25 Hz = $4e-6$ at -160 deg. Give V/F and A/F (mag + phase). [A: $\omega=157$: $V=6.28e-4$ at -70 ; $A=0.0986$ at $+20$]
- * **P5.** 2x2 rig EX-M: verify $k_2 = \omega^2 m_2$ numerically. [A: $261.8 \times 1.382 = 361.8 = \omega^2(T) m_2$. Done in text - redo blind!]
- * **P6.** Why does h12 mode-2 imag peak point down in EX-M? [A: $A_{12,2} = -0.447 < 0$: shapes oppose (1×-0.618). Sign, not error.]
- * **P7.** Your fit gives $f_1 = 50.1$ Hz from h11 but 51.3 from h22. Verdict? [A: Reject - poles are GLOBAL. Widen band, fix count, refit globally.]
- * **P8.** Blade to 50 Hz keeps 3 modes; statics off 8%. Fix? [A: Add residual stiffness (upper) + check lower out-of-band; do NOT add fake modes.]



(Revision) One-page recap

#	Big idea	Punchline
1	Pole anatomy	Real = damping, imag = freq; pairs
2	S-plane map	Damping pushes west on circle; $r = \omega_n$
3	Resonance signature	90 deg lag; K/x cancel; damping resists
4	Damping quickies	$z \sim D\omega/2\omega_n$; $d \sim 2\pi z$; pros fit
5	3 costumes	Polynomial / pole-zero / partial fractions
6	Residue	Pole strength; pole+residue = DNA
7	FRF = slice	$H(j\omega)$ cut along $j\omega$ axis
8	3 portraits	Bode / co-quad / Nyquist circle
9	Regions	Low stiff, reso damp, high mass
10	D/V/A slopes	(0,-2)/(+1,-1)/(+2,0): ID any plot
11	Eigensolution	det $\rightarrow$ freqs $\rightarrow$ shapes; equilib.+nodes
12	Uncouple	$x = [U]p$; m SDOFs replace n ($m \ll n$)
13	Global poles	Same denom; residues=shapes+reciprocity
14	FRF sum	$h_{ij} = \text{SUM shape} \cdot x \cdot \text{shape} \cdot x \cdot \text{SDOF}$
15	Full loop	FEM $\rightarrow$ predict vs test $\rightarrow$ extract; same objects
16	Masterclass receipt	2DOF statics matched to 4 decimals

$$\omega_n = \sqrt{k/m} \quad - \quad s = -z\omega_n \pm j\omega_d \quad - \quad z \sim D\omega/2\omega_n \quad - \quad ([K] - \omega^2[M])u = 0 \quad - \quad x = [U]p$$

(Self-test) 12 questions – no peeking!

- * Q1. Pole at $-3 \pm j40$. Give z , ω_n , ω_d . Light or heavy?
- * Q2. Why does the resonance peak measure damping and nothing else?
- * Q3. Half-power at 49.5/50.5 Hz, peak 50 Hz. Estimate z and Q .
- * Q4. Real part zero + imaginary peak at some frequency. What is it?
- * Q5. Nyquist looks like a potato. List 3 suspects.
- * Q6. Unlabeled FRF: flat low end, -2 slope high end. Which form? Get V/F ?
- * Q7. Why are poles 'global' but residues 'local'?
- * Q8. Reference at a mode's node kills it – prove in one line from hij.
- * Q9. 200k-DOF FEM, response to 100 Hz, 12 modes below. How many equations? Rest?
- * Q10. FEM says 88 Hz, test says 82 Hz. One modeling + one test suspect?
- * Q11. D/F peaks below f_n but A/F peaks above – contradiction?
- * Q12. Fit gives different f_1 per FRF. Accept? Fix?

Answers (really no peeking!):

- A1. $\omega_d=40$, $z \approx 3 \rightarrow \omega_n \approx 40.1$, $z \approx 7.5\%$ – moderate.
- A2. At ω_n , elastic/inertial forces cancel exactly; only $c \dot{x}$ opposes F .
- A3. $z \approx (50.5-49.5)/(2 \cdot 50) = 1\%$; $Q \approx 50$.
- A4. Resonance on a co-quad plot (clean, separated mode).
- A5. Overlapping modes, noise/leakage, nonlinearity (or calibration).
- A6. D/F compliance. Multiply by $j\omega$.
- A7. Poles = denominator roots, same all i,j ; residues = numerator, per pair.
- A8. $u_{jr} = 0$ at node $\rightarrow$ every $u_{ir} \times u_{jr} = 0$ for mode r . QED.
- A9. 12 modal SDOFs; higher modes quasi-static via residuals.
- A10. Modeling: wrong BCs / M-K layout. Test: mass loading, calibration, temperature.
- A11. No: D peaks below ω_n , V at ω_n , A above (three-peaks nuance). Say which FRF!
- A12. Reject: poles are global. Widen band, fix mode count, global refit.

Theory-to-lab checklist:

Before: predict band + mode count? reference off expected nodes? A/F or V/F – slope-check first plot?

During: 90 deg at every claimed resonance? Nyquist circles? drive-point alternation? coherence ~ 1 ?

After: poles stable across rows/columns (global!)? shapes orthogonal-ish? FEM updated + re-compared?



(Glossary) Ch.2 definitions XXL

Pole - root of char. equation; $-z \omega_n \pm j \omega_d$; damping + freq.

Residue - pole's finite strength; MDOF: shape- i x shape- j .

Zero - numerator root; drive-point zeros = anti-resonances.

S-plane - real-vs-imag pole map; $j\omega$ axis hosts FRFs.

Dynamic amplification - X/X_{static} vs b ; peak $\sim 1/(2\zeta) = Q$.

Phase lag - force $\rightarrow$ response: $0 \rightarrow 90$ (reso) $\rightarrow 180$ deg.

Half-power / log-dec - freq-/time-domain damping quickies.

Q factor - $1/(2\zeta)$; peak gain; sharpness.

Transfer function $H(s)$ - response/input in Laplace land; S-plane surface.

Partial fractions - $H(s)$ as SUM residue/(s-pole); test-ready form.

Cover-up - $A = \lim_{s \rightarrow p} (s-p)H(s)$: residue limit trick.

Bode / co-quad / Nyquist - mag-phase / re-im-vs-f / im-vs-re portraits.

Compliance/mobility/inertance - D/F, V/F, A/F (+ stiffness/impedance/mass inverses).

Stiffness/damping/mass regions - low-f / reso / high-f FRF rulers.

Eigenvalue/vector/pair - ω^2 / shape / the pair; $\det([K]-L[M])=0$.

Orthogonality - shapes decouple M and K; gives modal m, k.

Modal matrix $[U]$ - shapes as columns; $x=[U]p$ uncouples.

Modal force - $u(T)F$: how hard input pushes each mode.

Modal superposition - solve tiny modal SDOFs, project back, add.

Mass normalization - $\Phi^T M \Phi = 1$: scale for correct levels.

Truncation + residuals - keep $m \ll n$ modes; fold rest quasi-statically.

System matrix $[B(s)]$ - s-domain coefficient matrix; symmetric.

Global vs local - poles shared; residues per i, j .

Reciprocity - $H_{ij} = H_{ji}$ from symmetry.

Pseudo-orthogonality - test-vs-FEM shapes check via mass matrix.

EMA loop - FEM $\rightarrow$ predict vs test $\rightarrow$ extract; same poles/residues.

Council sign-off: Meera (theory) ok - Viktor (lab) ok - Lena (signals) ok - Arjun (pedagogy) ok - Sofia (field) ok.

Extended edition: every page earned. Next: Ch.3 signals!